
SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

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ABSTRACT

Spatial reasoning constitutes a core component of human cognition, requiring the representation of spatial layouts, relations, and transformations; however, it remains a persistent limitation for contemporary vision–language models (VLMs). Existing evaluations have primarily used synthetic or LLM-generated environments, constrained task schemas, and puzzle-like problem formulations, which do not adequately instantiate the visual clutter, perceptual ambiguity, and heterogeneous spatial relations present in natural scenes. To address this measurement gap, we introduce **SPATIALAB**, a benchmark designed to evaluate VLM spatial reasoning under realistic and unconstrained visual conditions. SPATIALAB contains 1,400 visual question–answer pairs organized into six primary categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*. Each primary category is further decomposed into five subcategories, producing 30 distinct task types. The benchmark includes at least 25 questions per subcategory and at least 200 questions per primary category, and it supports evaluation in both multiple-choice and open-ended response formats. We evaluate a broad set of state-of-the-art VLMs, including open-source models, closed-source proprietary models, reasoning-oriented models, and models specialized for spatial reasoning. The results identify a substantial gap between model performance and human performance. In the multiple-choice setting, InternVL3.5-72B obtains 54.93% accuracy, whereas humans obtain 87.57%. In the open-ended setting, all models exhibit an additional performance reduction of approximately 10–25%, with GPT-5-mini achieving the best model score of 40.93% compared with 64.93% for humans. These outcomes indicate systematic limitations in reasoning over complex spatial relations, depth and occlusion cues, navigation constraints, and 3D geometric structure. By specifying a diverse real-world evaluation protocol, SPATIALAB exposes concrete failure modes in current VLM spatial competence and provides a diagnostic benchmark for future work on robust, human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning is a foundational component of human cognition that supports the construction of spatial layouts, mental transformation and manipulation of objects, and navigation through physical environments. It is required in application domains including robotics, autonomous driving, augmented and virtual reality, geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For artificial agents that perceive and act in dynamic environments, reliable spatial competence requires the joint interpretation of geometric relations, depth cues, occlusion patterns, and egocentric viewpoints in scenes that are visually noisy and structurally complex (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Although recent vision–language models have improved multimodal representation learning and language grounding, their spatial judgments remain unstable in realistic visual conditions (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025).

Existing spatial benchmarks remain narrow and simplified. Many emphasize elementary tasks, including binary spatial relations, low-resolution depth labels, or schematic navigation cues, and often use synthetic or simplified settings that reduce clutter and homogenize object appearance (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024;

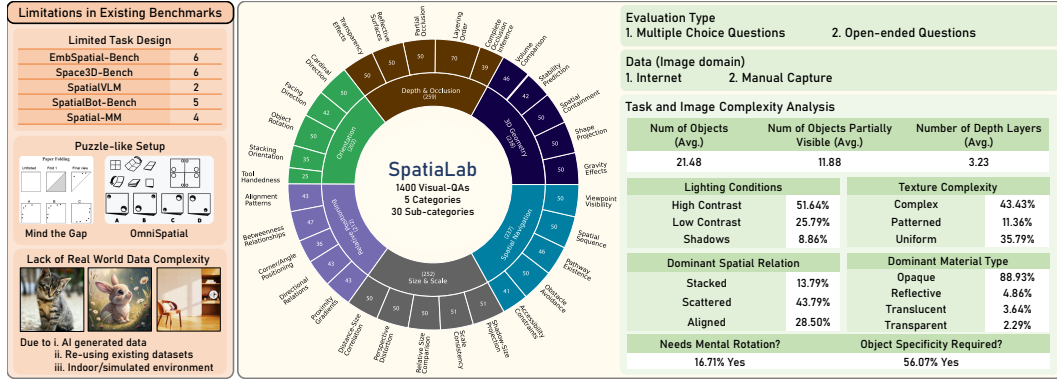


Figure 1: **Overview of SPATIALLAB.** The benchmark addresses limitations of prior datasets, introduces 1,400 visual QA pairs across the spatial taxonomy, and supports systematic evaluation with multiple-choice and open-ended tasks. The images instantiate diverse complexity factors, including object count, depth layers, lighting, textures, relations, and materials.

Song et al., 2024; Yang et al., 2024; Qi et al., 2025; Cai et al., 2024). Such controlled setups reduce perceptual and inferential load, which can produce apparent saturation while masking brittleness under distribution shift. Critical capabilities, including robust occlusion inference, scale-consistent judgment across viewpoints, and planning under partial observability, remain under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). Consequently, models that perform well on existing benchmarks can still fail on integrated, multi-step spatial inference, which is problematic for real-world deployment where spatial errors have practical consequences.

A meaningful evaluation must span the core axes of spatial reasoning: relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry. Human spatial cognition integrates these dimensions under noise, occlusion, and ambiguity, whereas task-isolated benchmarks often evaluate them independently (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). A robust benchmark should therefore combine selection and generation formats, compare against human performance, and use real-world scenes that expose failures hidden by clean simulator datasets; these principles motivate SPATIALLAB.

To provide a more discriminative evaluation, we introduce **SPATIALLAB**, a benchmark for spatial reasoning in real-world images, as outlined in Figure 1. SPATIALLAB contains 1,400 visual question-answer pairs collected from natural scenes and organized into six major categories: relative positioning, depth and occlusion, orientation, size and scale, spatial navigation, and 3D geometry. Each category is decomposed into five fine-grained subcategories, yielding 30 task types that cover both elementary relational judgments and higher-order spatial inference. Each item is available in a multiple-choice format and an open-ended format, enabling controlled measurement of recognition-oriented selection and free-form generative reasoning.

We evaluate a diverse set of contemporary VLMs, including proprietary systems, open-source models at multiple scales, reasoning-oriented models, and spatially specialized models. The empirical results show that current models remain far below human performance on SPATIALLAB. The best multiple-choice model, InternVL3.5-72B, reaches 54.93% accuracy, while humans achieve 87.57%. In open-ended evaluation, all models incur an additional 10–25% reduction, and the best model, GPT-5-mini, reaches 40.93% compared with the human score of 64.93%. These discrepancies indicate that stronger generic VLM capability and larger parameter count do not automatically yield robust spatial inference under natural image complexity.

Our analysis further localizes model failures to specific spatial mechanisms. Models exhibit persistent errors in occlusion reasoning, depth ordering, viewpoint-dependent orientation, scale consistency, navigation feasibility, and 3D geometric inference. We also test performance-improvement strategies, including chain-of-thought prompting, self-reflection, supervised fine-tuning, and agentic decomposition. These interventions provide measurable but limited gains, suggesting that the remaining gap is not primarily a prompt-format artifact but reflects deeper limitations in the learned visual-spatial representations and inference procedures.

Table 1: Technical comparison of spatial reasoning benchmarks by task setting, data properties, evaluation format, and reported baseline performance.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval			A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.		Type	Depth					
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I	I	-	-
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I	I	LXMERT	70.10
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I	I	XVLM-COCO	60.40
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I	I	Qwen-VL-Max	49.11
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I	I	RAG3D-Chat	66.80
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I	I	GPT-4o	57.83
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I	I	GPT-4o	59.03
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I	I	GPT-4 Vision	63.82
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I	I	LEO	71.90
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'25C	Open	L	S	I	I	GPT-4V	68.00
VSI-Bench	Y	8	Indoor	M,E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V	I	Gemini-1.5 Pro	45.40
OmniSpatial	Y	50	Internet	Internet	M,P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I	I	o3-2025-04-16	56.33
Mind the Gap	N	6	Mix	AI-Gen	M,P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I	I	InternVL2.5-26B	48.83
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix	I	Gemini-2.5 Pro	62.00
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix	I	InternVL3.5-72B	54.93
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix	I	GPT-5-mini	40.93

Our contributions are threefold. First, we construct **SPATIALAB**, a real-world spatial reasoning benchmark with 1,400 questions spanning six categories and 30 subcategories. Second, we provide paired multiple-choice and open-ended evaluation protocols that reveal both recognition-level and generative reasoning failures. Third, we conduct a broad empirical assessment across state-of-the-art VLM families and analyze failures, performance gaps, and improvement strategies, thereby providing a structured diagnostic resource for future work on spatially grounded multimodal intelligence.

2 PRELIMINARIES: SPATIAL REASONING IN VLMs

Visual-spatial reasoning denotes the capacity to perceive, encode, and transform spatial relations among objects, and it is central both to human cognition and to embodied agents that must navigate and act in physical environments (Palmer, 1999; Chabris et al., 2006). Formally, a scene may be represented as a set of objects $O = \{o_1, \dots, o_n\}$ with attributes such as position, orientation, and size defined in a world coordinate system W . Visual-spatial reasoning can then be formulated as learning a mapping $f : (I, Q) \mapsto R$, where I denotes an image or image sequence, Q denotes a linguistic query, and R is a structured relation set $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ containing predicates such as *left-of*, *occludes*, or *supports*. For vision-language models (VLMs), estimating f is difficult because the model must couple noisy visual evidence with compositional linguistic abstraction while respecting embodied constraints such as perspective, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022).

Earlier VLM evaluations of spatial reasoning typically instantiate basic relation labels or narrowly specified task schemas (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024). Many such evaluations are based on synthetic datasets such as CLEVR (Johnson et al., 2016), structured question-answering benchmarks such as GQA (Hudson & Manning, 2019), or puzzle-like simulated settings such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis et al., 2025), as summarized in Table 1¹. Although these resources support controlled measurement, they simplify spatial structure, suppress environmental noise, and underrepresent factors such as temporal variation and viewpoint change. A more valid benchmark should therefore instantiate spatial reasoning in real-world contexts with heterogeneous scene structure, multiple perspectives, and perceptual ambiguity. These conditions better approximate the spatial demands encountered in embodied and interactive applications, including robotics, AR/VR, and autonomous navigation (Mu et al., 2023; Collaboration et al., 2023).

In **SPATIALAB**, spatial reasoning is organized into six principal categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*. Each category corresponds to a distinct spatial-cognitive faculty studied in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, each category is decomposed into five subcategories, yielding a taxonomy of thirty task types that isolate fine-grained reasoning operations such as directional alignment, occlusion inference, and stability prediction. This taxonomy converts broad spatial competence into measurable units, enabling analysis of which dimensions

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

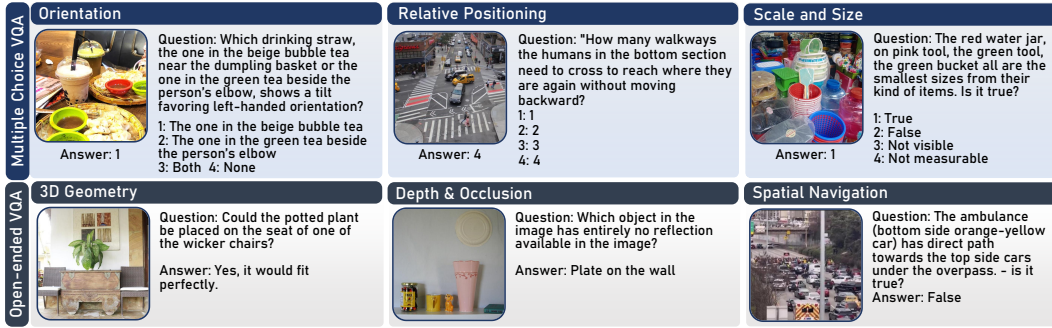


Figure 2: Representative SPATIALAB examples from the six primary spatial categories, shown in both open-ended and MCQ formats.

are solved reliably and which remain fragile or underdeveloped (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

The benchmark uses two complementary response formats: multiple-choice and open-ended question answering. The multiple-choice format measures discriminative selection among controlled distractors, whereas the open-ended format measures generative spatial reasoning and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Jointly, the two formats separate answer-recognition effects from free-form spatial inference and expose different failure modes in model behavior. By combining a principled taxonomy, diverse real-world imagery, and dual evaluation protocols, SPATIALAB provides a structured framework for systematic benchmarking, fine-grained model comparison, human-model alignment analysis, and diagnosis of robust spatial cognition in artificial agents.

3 SPATIALAB BENCHMARK

Spatial reasoning remains difficult for vision-language models (VLMs), particularly in naturalistic images where occlusion, perspective distortion, clutter, and background noise co-occur. We introduce **SPATIALAB** as a benchmark for measuring complex spatial cognition under realistic visual conditions. The benchmark contains 1,400 visual question-answer pairs distributed across six primary categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*. Each primary category is partitioned into five subcategories, producing 30 task types. Each subcategory contains at least 25 QA pairs, and each primary category contributes at least 200 QA pairs, which provides balanced coverage across the taxonomy. **SPATIALAB** supports both multiple-choice and open-ended evaluation, thereby measuring discriminative answer selection as well as generative spatial reasoning and explanation. The combination of diverse imagery, fine-grained annotation, and multi-stage quality control makes **SPATIALAB** a scalable diagnostic resource for real-world spatial reasoning; representative instances are shown in Figure 2.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The image corpus was constructed using three complementary acquisition strategies: (i) automated large-scale web crawling, (ii) targeted online retrieval for scene-specific visual configurations, and (iii) manual snapshots captured in diverse indoor and outdoor environments (Figure 3). This multi-source pipeline increases coverage over object categories, environmental conditions, and spatial configurations. To characterize natural variability explicitly, each image was profiled along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). These dimensions were represented in the final corpus so that the benchmark reflects in-the-wild visual variation rather than simplified synthetic regularities.

Corpus-level complexity statistics indicate that each image contains 21.48 objects on average, including 11.88 partially visible objects, distributed across 3.23 depth layers. Spatial reference chains, such

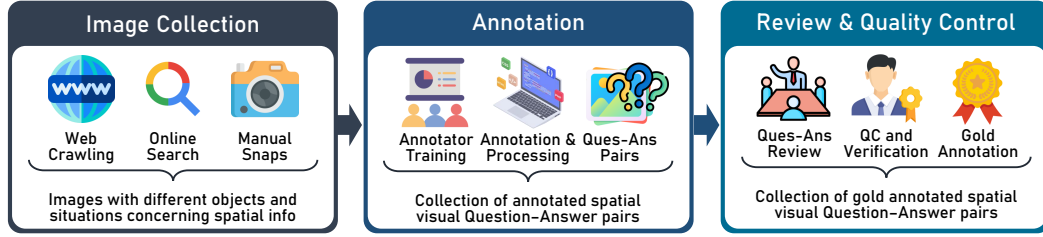


Figure 3: **Data creation pipeline for SPATIALAB.** Images are acquired through web crawling, targeted search, and manual snapshots, then converted into structured spatial question-answer pairs through annotation, review, and validation.

as “object A is left of B, which is behind C,” contain 2.07 links on average, indicating that many items require multi-step relational inference. In addition, 16.71% of questions require mental rotation, and 56.07% require object-specific grounding through color, pattern, or other visual details. These measurements show that **SPATIALAB** encodes cluttered, layered, and relationally dense scenes that are characteristic of real-world spatial reasoning and underrepresented in synthetic datasets.

Annotation. Annotation was performed in three phases. First, annotators completed targeted training, with details provided in Appendix C.2, to standardize interpretation of the spatial categories and subcategories. Second, each image was associated with one or more spatial QA tasks covering reasoning types such as proximity gradients, alignment patterns, occlusion inference, and stability prediction. The questions were designed to require both perceptual grounding and higher-order inference, for example predicting stability under gravity constraints. Third, each QA pair was encoded in both multiple-choice (04 options) and open-ended formats to support direct comparison across evaluation paradigms. This process produced approximately 1,500 spatial visual QA pairs across six categories and 30 subcategories before final validation.

Review and Quality Control. Annotation outputs were subjected to a three-tier review and validation pipeline. The first stage checked semantic validity and alignment with the assigned subcategory. The second stage used independent annotators to verify the question wording and designated answer, with emphasis on eliminating ambiguity and enforcing task clarity. The third stage established a gold-standard annotation round consisting of high-confidence QA pairs for evaluation. Image metadata was removed, and the dataset was curated with attention to copyright constraints, although minor residual oversights may remain. The pipeline yielded a final benchmark of 1,400 validated QA items.

4 EXPERIMENTS AND EVALUATION

Models. The evaluation suite is organized into four model groups. **Proprietary models** include GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** include Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). **Reasoning models** include o4-mini (OpenAI, 2025), Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** include SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE, we report only open-ended evaluation because these models are not instruction-tuned for the MCQ interface and were trained primarily on open-ended tasks, which produced excessive refusals under MCQ prompting. For readability in the result tables, model groups are color coded as follows: proprietary, open-source up to 7B, larger than 7B, proprietary reasoning, open-source reasoning, spatial specialists, and human baseline.

Evaluation Process and Metrics. Accuracy is the primary metric for both MCQ and open-ended tasks. In **MCQ evaluation**, models are prompted directly to return the option number corresponding to the selected answer, and responses are automatically matched against the reference option. The

Table 2: Multiple-choice evaluation accuracy (%) (↑) on SPATIALAB-MCQ, disaggregated by primary question category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

complete prompts and implementation details are provided in Appendix D.1. In **open-ended evaluation**, models generate free-form answers, which are scored by an LLM-as-a-judge protocol; the corresponding prompts and procedural details are provided in Appendix D.2. To validate the judge, we perform an LLM–human agreement analysis as described in Appendix D.2. Table 5 shows that the LLM judge (Gemini-2.5-Flash) attains substantial agreement with human annotators, with Cohen’s kappa of 0.738 against majority vote and 0.681–0.795 against individual annotators. Raw accuracy against majority vote is 0.880, and Fleiss’ kappa among human annotators is 0.774, indicating strong inter-annotator reliability. Additional experiment settings, prompts, and evaluation procedures are reported in Appendix D.

Improving Visual Reasoning Capabilities. We examine several mechanisms for improving spatial reasoning on SPATIALAB: inherent reasoning in VLMs (Appendix H.1), Chain-of-Thoughts (CoT) prompting (Appendix H.2), CoT with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and multi-agent systems (Appendix H.5). For SFT, Qwen-VL2.5-3B-Instruct is fine-tuned on a stratified 40% split of the dataset and evaluated on the remaining 60%. This procedure improves generalization on relational and geometric tasks and reduces errors in navigation and orientation, with additional gains when CoT and SFT are combined. However, the agentic setting shows category-dependent behavior: orientation improves substantially, whereas other categories stagnate or degrade. This pattern indicates that multi-step reasoning procedures do not substitute for stronger perceptual and spatial grounding.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. Table 2 reports multiple-choice performance on SPATIALAB-MCQ. Model accuracy varies substantially across model families, task categories, and individual systems: overall accuracies range from approximately 30% to 55%, compared with a random-choice baseline of 25% and a human baseline of 87.57%. The highest-scoring model is the open-source InternVL3.5-72B, which obtains 54.93% accuracy. Reasoning- and instruction-tuned models, including o4-mini

Table 3: Open-ended evaluation accuracy (%) (↑) on SPATIALAB-OPEN, disaggregated by primary question category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline							
	73.53	50.19	70.30	69.81	65.48	62.87	64.93

(53.21%), Gemini-2.5-Flash-Thinking (52.93%), and GPT-5-mini (54.29%), and several large open-source models, including InternVL3.5-72B (54.93%) and Llama-3.2-90B (50.36%), occupy the top performance range. However, parameter scale is not sufficient: Llama-3.2-11B obtains only 30.50% despite its substantial size. At the category level, Orientation and 3D Geometry are comparatively stronger for many high-performing models, with several models exceeding 60% in at least one of these categories, whereas Spatial Navigation, Depth & Occlusion, and Size & Scale are more variable and generally more difficult. The spatially specialized models SpaceOm, SpaceThinker, and SpaceQwen achieve only mid-40s to low-40s overall accuracy, indicating that specialization does not automatically transfer to broad MCQ competence across this taxonomy. The mixed outcomes in Depth & Occlusion and Size & Scale suggest that architectures differ in their inductive biases and training exposure for occlusion, depth, and scale cues. Overall, instruction tuning and explicit reasoning appear to support geometric and orientation judgments, while multi-step grounded tasks such as navigation remain a persistent weakness.

SPATIALAB-OPEN. Table 3 shows that open-ended evaluation produces substantially lower scores than MCQ evaluation. Overall accuracy ranges from approximately 9.6% for InternVL3.5-1B to 40.9% for GPT-5-mini, while the human baseline is 64.9%. Proprietary systems generally lead the open-ended setting, including GPT-5-mini (40.93%), GPT-4o-mini (26.00%), and Gemini-2.5-Flash (30.93%). Reasoning-tuned systems also perform strongly relative to many open-source baselines: o4-mini reaches 37.86%, and Gemini-2.5-Flash-Thinking reaches 32.77%, suggesting that instruction and CoT-style tuning improve generative spatial responses. Most small open-source models cluster at low accuracy levels, often in the teens, including InternVL3.5-1B at 9.64% and Qwen-VL2.5-3B at 12.93%. Larger open models improve only modestly into the low-20s, for example InternVL3.5-72B at 23.36% and Qwen-VL2.5-72B at 24.64%. Spatial-specialist models achieve approximately 10–13% overall accuracy, indicating that their specialized design does not yield robust open-ended generation across the benchmark. Across subtasks, top models perform relatively better on orientation and selected relative-positioning items, such as GPT-5-mini with 49.53% on Rel.Pos and 37.13% on Ori, whereas depth and occlusion, size and scale, and especially

spatial navigation remain consistently difficult, with many models below <30% on these subtasks. These results identify multi-step grounding and perceptual inference as the dominant bottlenecks for open generative spatial answering.

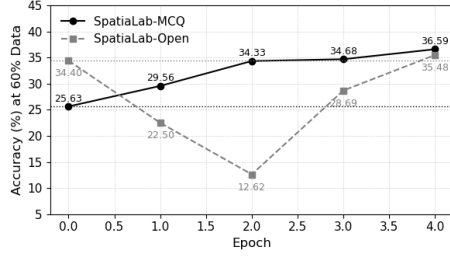
5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

Across 25 models, the average MCQ→open-ended performance gap on SPATIALAB is 23.0% ($\sigma = 5.5\%$), with subtask means spanning 22.89% for spatial navigation to 24.57% for 3D geometry. Specialist spatial reasoning models exhibit the largest gaps, approximately 27%, with particularly large disparities in spatial navigation (up to 36.68%) and orientation (34.44%). In contrast, reasoning-oriented models show smaller gaps of approximately 19% and lower subtask variance, consistent with a stabilizing effect from instruction tuning and CoT decoding. Correlation analysis indicates that spatial navigation dominates the aggregate disparity (Pearson $r = 0.99$), while orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also contribute. Negative or near-zero gaps, such as Llama-3.2-11B at -1.98 in depth & occlusion and o4-mini at -3.18 in relative position, indicate that MCQ distractor design can sometimes misrepresent model competence. We hypothesize that format sensitivity arises from the structural advantage of MCQ answer selection, specialization bias toward categorical response formats, sequential reasoning demands in spatial navigation, and calibration differences in generative output. Therefore, MCQ-only evaluation can overestimate practical spatial reasoning ability. We recommend pairing MCQ with open-ended evaluation, auditing distractors, using stepwise generation or instruction tuning where appropriate, and reporting per-subtask diagnostics to reduce format-dependent artifacts. Additional details are provided in Appendix G and Table 18.

5.3 ERROR ANALYSIS

SPATIALAB-MCQ. The MCQ error analysis (Appendix E) shows that VLMs can achieve localized high scores while lacking holistic spatial competence. Closed-source models reach high performance on selected subcategories, for example 85.71% in *Stacking Orientation*, but collapse to near chance on tasks such as *Relative Size Comparison*. Open-source scaling raises some ceilings, including 80.56% in *Corner/Angle Positioning*, but does not prevent severe failures such as 2.0% in *Object Rotation*. Reasoning-augmented and spatially specialized models provide localized gains in areas such as occlusion inference or navigation, but remain below 55% on several physical abstraction tasks, including *Gravity Effects* and *Stability Prediction*. The error distributions identify systematic weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local visual cues (*Reflective Surfaces*). Higher scores in *Obstacle Avoidance* are consistent with possible shortcut exploitation rather than robust path planning. These patterns indicate reliance on surface-level correlations and insufficiently stable encodings of orientation, physics, and compositional spatial logic. Progress likely requires richer training distributions, such as embodied and physics-driven data, together with architectural mechanisms for geometric and reference-frame grounding.

SPATIALAB-OPEN. The open-ended error analysis (Appendix F) shows that even top-performing closed-source models have task-specific rather than global competence. GPT-5-mini reaches 58.14% in *Directional Relations*, but fails on tasks such as *Proximity Gradients*. Large open-source models obtain strong results in *Depth & Occlusion*, including 60.0%, but still fall to 9.3% on *Proximity*, indicating that scale alone does not ensure robustness. Smaller models show isolated strengths, such as 26.0% in *Relative Size Comparison*, but often fail completely, including 0.0% in *Betweenness*. Reasoning-tuned systems provide evidence that explicit reasoning can help, as Gemini-2.5-Flash-Thinking reaches 75.0% in *Tool Handedness*; however, related systems fall to single-digit scores, revealing fragile integration. Specialized spatial models perform worst overall, rarely exceeding 20% and frequently failing on core relational tasks. The failure patterns emphasize weaknesses in occlusion handling, orientation, and multi-step relational chaining, while partial success on size-scale cues suggests dependence on superficial correlations. The results point to three root causes: missing representations for spatial analogies, architectural limits associated with scale and training diversity, and brittle reasoning pipelines. Addressing these limitations will require unified geometric encodings, physics-aware reasoning, and embodied data to move from isolated performance peaks toward consistent spatial competence.



Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(a) Supervised fine-tuning performance trajectories across epochs. (b) Category-level performance comparison before and after supervised fine-tuning for MCQ and open-ended formats.

Figure 4: Supervised fine-tuning results: (a) learning trajectories over epochs and (b) final accuracy values.

Qualitative Error Analysis. The qualitative error analysis in Appendix I indicates that model failures cluster into a small set of recurring categories rather than appearing as random noise. Common failures include spatial mislocalization, in which models select the wrong referent in crowded scenes; perspective and scale errors, in which object-size priors override image-specific evidence; and occlusion or ordering errors, especially for thin or partially hidden structures. Additional failures include attribute confusion, where perceptual properties are conflated with functional properties, and open-ended rationalizations that are linguistically fluent but visually ungrounded. These errors recur across architectures and prompting strategies, suggesting structural biases in current VLMs rather than benchmark-specific artifacts. Failure rates increase when multiple cues must be fused, such as depth ordering with relative size, and confidence calibration is especially weak in open-ended settings. The diagnostic protocols show that models often ignore minimal but decisive visual features, rely on brittle heuristics, and fail to update predictions under counterfactual edits. These patterns indicate insufficient object-centric binding, limited geometric supervision, and training objectives that reward plausibility over grounding. Consequently, current VLMs can support coarse perception but remain unreliable for multi-cue integration and grounded spatial reasoning, motivating geometry-aware supervision, multi-scale feature retention, and verification pipelines.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. Reasoning-enabled models consistently outperform their corresponding baselines in both MCQ and open-ended formats, with the largest gains in relational and orientation tasks, such as +13.1% in Relative Positioning for MCQ. Reasoning mechanisms stabilize open-ended performance, but improvements remain non-uniform across categories and decline in Size & Scale. This pattern indicates that reasoning modules improve logical consistency while leaving perceptual grounding and scale sensitivity only partially addressed. Detailed per-category comparisons are provided in Appendix H.1.

Chain-of-Thought (CoT) Prompting. Across models, CoT prompting provides limited benefit and often reduces accuracy (Table 19), with Orientation as the only category showing consistent improvement. This result suggests that step-by-step verbal reasoning can assist directional alignment, but does not resolve tasks that require robust perceptual grounding, including depth, scale, and spatial navigation. Unlike many text-only reasoning settings, CoT in SPATIALAB can amplify incorrect priors rather than correct them, indicating that the benchmark requires perceptual understanding beyond logical chaining. Additional details are provided in Appendix H.2.

Chain-of-Thought (CoT) with Self-Reflection. Adding self-reflection to CoT prompting, where models are instructed to review their previous reasoning step by step and correct errors (see Figure 8 in Appendix for the prompt), yields modest gains in multiple-choice settings, especially for geometry and depth. However, the gains do not transfer reliably to open-ended tasks (Appendix H.3). This outcome indicates that reflective prompting can improve answer selection under constrained formats but cannot compensate for missing perceptual grounding, which remains a core limitation of current VLMs.

Supervised Fine-Tuning (SFT). As shown in Figure 4, supervised fine-tuning (SFT) on Qwen2.5-VL-3B-Instruct using a stratified 40% split of SPATIALAB (~560 samples) consistently improves MCQ accuracy across all spatial reasoning categories. However, transfer to open-ended tasks is limited and can be negative. This divergence is consistent with overfitting to

task-specific answer distributions and with the concern that SFT can *teach to the test* rather than induce generalizable reasoning (Wang et al., 2025). The open-ended performance drop suggests that the fine-tuned model develops representations optimized for categorical discrimination but unstable for generative spatial language. A likely contributor is catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a), combined with objectives that favor discrete-choice alignment over spatial explanation. Thus, although SFT improves structured-format accuracy, it can constrain the model toward benchmark-specific response patterns and impede progress on naturalistic spatial reasoning. Additional details are provided in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). To evaluate agentic decomposition, we use SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025) for structured spatial reasoning over images. The framework decomposes complex visual reasoning into specialized sub-tasks handled by dedicated agents, including object segmentation, attribute extraction, spatial relation mapping, and transformation tracking. The intermediate outputs are consolidated into a unified structured representation intended to propagate low-level perceptual evidence into higher-order relational inference. We implemented the pipeline with Gemini-2.5-Flash at low temperature to produce deterministic outputs during multi-stage refinement. The results show substantial gains in Orientation, with +8.00% in MCQ and +36.00% in open-ended evaluation. However, the improvement is category-specific: Depth & Occlusion and Spatial Navigation decline sharply by -24.00% and -12.00% in open evaluation, respectively. These declines indicate that when core perceptual priors such as depth and occlusion are weak, multi-step reasoning can reinforce erroneous intermediate assumptions rather than correct them. Agentic workflows therefore help tasks reducible to sequential alignment but do not compensate for insufficient perceptual grounding in complex 3D environments. Additional details are provided in Appendix H.5.

6 DISCUSSION

The empirical results indicate that spatial reasoning in contemporary VLMs remains weak when the evaluation requires integration of natural-image perception with relational, geometric, and navigational inference. Performance degradation is especially pronounced in open-ended evaluation, where models must produce the relevant spatial relation without support from enumerated alternatives. This gap suggests that multiple-choice accuracy can overestimate spatial competence by permitting answer elimination and lexical matching rather than requiring explicit construction of a spatial representation.

The error analyses further show that failures are not uniformly distributed across the taxonomy. Depth and occlusion, 3D geometry, scale consistency, and multi-step navigation require models to infer latent scene structure from incomplete or viewpoint-dependent visual evidence. Current models often substitute local object recognition, saliency cues, or language priors for this latent spatial inference. As a result, even models with strong general multimodal performance can fail on questions that require maintaining consistent object relations across depth, orientation, and viewpoint transformations.

The improvement experiments show that prompting, self-reflection, supervised fine-tuning, and agentic decomposition can increase accuracy but do not remove the human-model gap. These findings imply that stronger reasoning scaffolds can help models expose and reuse relevant cues, but they cannot fully compensate for insufficient geometric grounding or unstable spatial representations. Future systems will likely require training signals and architectures that couple visual perception to explicit relational structure, action-conditioned feedback, and multi-scale geometric reasoning.

7 CONCLUDING REMARKS

Spatial reasoning remains a central bottleneck for VLMs, particularly in visually complex real-world settings. SPATIALAB specifies an evaluation benchmark with 1,400 diverse QA pairs distributed across six spatial categories and thirty subcategories, and it supports fine-grained assessment under both MCQ and open-ended response formats. Across multiple model families and performance-improvement procedures, the experiments reveal large gaps between state-of-the-art models and human performance; open-ended generation additionally produces a 10–25% reduction relative to multiple-choice accuracy. The error analyses identify persistent failures in depth inference, occlusion

reasoning, 3D geometry, and multi-step navigation, indicating that many models rely on superficial visual or linguistic cues rather than stable grounded spatial representations. **SPATIALAB** therefore functions as a structured diagnostic testbed for measuring spatial competence under natural-image conditions.

The benchmark enables holistic evaluation and provides actionable evidence for developing VLMs with human-aligned spatial understanding. Progress will require richer supervision signals, such as embodied interaction and action-conditioned feedback, together with architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

We implemented multiple controls to maintain ethical compliance during dataset construction and evaluation. Image-associated metadata was removed to protect privacy, and personally identifiable visual features, including faces and license plates, were minimized or excluded whenever feasible. The dataset was curated with attention to copyright, source provenance, and licensing constraints, although minor residual oversights may remain despite these safeguards. The benchmark construction protocol was designed to be transparent and fair, with no demographic group targeted or negatively represented. We adhered to the ICLR Code of Ethics and confirm that the work does not involve human subjects or sensitive personal data.

REPRODUCIBILITY STATEMENT

To support reproducibility, the main text and appendix specify the dataset construction process, annotation pipeline, and evaluation protocol. Filtering, quality-control, and validation procedures are documented explicitly, including definitions for task categories and annotation criteria. The appendix also reports hyperparameters, model configurations, and training procedures, enabling consistent replication of the reported results.

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A RELATED WORK

Spatial reasoning in multimodal models. Spatial reasoning denotes the ability to represent geometric structure and relational layout in visual scenes, and it is a prerequisite for embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) evaluated compositional reasoning through synthetic or curated templates, enabling controlled analysis but limiting ecological validity. With the emergence of large vision–language models (VLMs) (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023), spatial reasoning has been evaluated in more naturalistic settings, yet systematic weaknesses persist. Models that perform well on captioning and visual question answering can still fail when tasks require depth ordering, occlusion inference, or consistent reference-frame alignment. This discrepancy indicates that parameter scaling and broad multimodal pretraining do not by themselves guarantee structured spatial understanding.

Spatial reasoning benchmarks. Recent benchmarks have been proposed to measure these limitations (Table 1). SCANQA (Azuma et al., 2022) was among the first embodied QA datasets requiring reasoning over 3D indoor environments, but its scale (800 scenes) and shallow task complexity restrict generalization. Datasets such as VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) extend evaluation to real-world or household contexts, but they rely on template-driven questions, which reduces linguistic variability and overrepresents simple binary or multiple-choice relations. More recent resources, including EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), increase task variety but remain constrained by scale (2.2K and 211 samples, respectively) and by annotation style. SPATIALRGPT-BENCH and BLINK-SPATIAL introduce mixed domains and up to 14 categories, improving diversity, but they still emphasize static snapshots or limited multi-view simulation.

Embodiment, difficulty, and diversity. The shift toward embodiment is visible in datasets such as ROBOSPATIAL (Song et al., 2024), which scales to 1M scenes with robot-centric questions. However, its template-based annotation reduces naturalness and can enable heuristic shortcuts. SPATIALVLM (Chen et al., 2024b) provides very large scale, with billions of QA pairs, but its two-category coverage is insufficient for fine-grained diagnostic evaluation. Conversely, high-quality manual datasets such as VSI-BENCH (Yang et al., 2024) improve annotation fidelity but contain only 288 scenes, limiting training utility and robust cross-model comparison. In terms of difficulty, most existing benchmarks remain at “medium” complexity (Mm in Table 1) and rarely instantiate “hard” or “extremely hard” tasks requiring multi-step causal inference. This leaves a substantial evaluation gap for next-generation multimodal systems.

Evaluation protocols. Evaluation format and analysis granularity are also critical. Most prior work uses closed-ended QA or binary judgments, as in VISUAL SPATIAL and EMBSPATIAL-BENCH; these formats simplify scoring but restrict output expressivity. Some datasets, including ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN, introduce open-ended evaluation, but only SPATIALAB-OPEN combines this format with high task complexity, manual puzzle-style annotation, and detailed error analysis. This combination supports analysis beyond aggregate accuracy, including calibration behavior and cross-model divergence. Without such granularity, apparent performance improvements can reflect dataset-specific shortcut learning rather than robust spatial competence.

Positioning of SPATIALAB. SPATIALAB addresses these limitations through broader taxonomy, real-world imagery, and dual-format evaluation. Whereas prior datasets typically span 2 to 14 task categories, SPATIALAB covers 30 subcategories across geometry, occlusion, orientation, relative positioning, size and scale, and navigation. Unlike template-based benchmarks, the questions are manually authored and puzzle-inspired, increasing linguistic richness and reducing shortcut opportunities. SPATIALAB is also embodied and multimodal, including static and dynamic visual contexts, thereby bridging robotics-style datasets such as ROBOSPATIAL and web-scale corpora such as SPATIALVLM. As shown in Table 1, even strong baselines such as InternVL3.5-72B and GPT-5-mini remain below 60%, and the open-ended variant reaches only 40.93% accuracy. This gap demonstrates both benchmark difficulty and diagnostic utility.

Toward next-generation evaluation. By combining high difficulty, broad taxonomic coverage, manual annotation, and open-ended evaluation, SPATIALAB extends spatial reasoning evaluation

from narrow template-driven settings to comprehensive embodied testing. The design supports not only binary correctness measurement but also systematic error analysis, causal and counterfactual probing, and calibration assessment. SPATIALAB therefore provides a rigorous basis for the next phase of multimodal evaluation, complementing datasets such as MIND THE GAP and VLM4D, which investigate temporal or generative aspects but have narrower spatial coverage. We position SPATIALAB as a high-bar diagnostic instrument for moving models beyond surface recognition toward grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To position SPATIALAB within the broader benchmark landscape, we examined representative spatial reasoning datasets. Existing resources provide important foundations, but they exhibit structural and methodological constraints that limit their diagnostic value for current VLMs. The main limitations are synthetic-domain bias, restricted task scope, and constrained evaluation formats. Table 4 summarizes these limitations.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA rely on synthetic environments or template-generated language. These settings provide controlled conditions for isolating compositional reasoning, but their visual domains omit real-world complexity and their linguistic space is narrow. Models can therefore exploit template regularities rather than perform grounded spatial interpretation. Simulated benchmarks such as ROBOSPATIAL face similar issues and can exhibit a substantial sim-to-real gap.

Domain and Scale Limitations. Datasets grounded in 3D reconstruction, including ScanQA and Space3D-Bench, provide depth supervision but are restricted to indoor scenes. This restriction prevents assessment in outdoor or mixed environments that are important for embodied agents. High-quality manual datasets such as VSI-Bench provide richer annotations but remain small, limiting high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM cover broad tasks but do not provide detailed taxonomies, making them less suitable for diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, including EMBSPATIAL-BENCH and WHAT’S UP, rely heavily on Binary or MCQ formats. These formats can inflate performance by enabling option elimination rather than requiring generative spatial reasoning. SPATIALAB mitigates these issues by combining MCQ and Open-Ended formats across a detailed taxonomy of 30 spatial subcategories.

Table 4: Limitations identified in representative prior spatial benchmarks compared with SPATIALAB. Limitations are categorized by domain constraints, evaluation format, and task taxonomy.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain; template-driven language; prone to shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Restricted to indoor point clouds; limited scale (< 1K scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a small number of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats; limited complexity in reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes produce a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~200 samples) prevents robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Emphasizes metric outputs; limited support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Addresses identified limitations with real-world images, detailed taxonomy, and diverse evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

Spatial reasoning requires more than object recognition; it requires representing how objects relate, align, occlude, support, and constrain one another in physical space. *SpatialLab* operationalizes this capability through fine-grained task families that evaluate structured spatial understanding across diverse visual contexts. The task set ranges from relative positioning, such as *What’s to the left of the microwave?*, to geometric and physical inference, such as *Would this stack fall if another book is added?*. Each primary category is decomposed into subcategories that isolate specific reasoning mechanisms, including occlusion, orientation, navigation, and gravity-mediated physical effects. This design requires models to move beyond object detection toward explicit spatial interpretation, and it supports evaluation in application contexts such as robotics, AR/VR, autonomous driving, and

human–computer interaction. The following sections define the rationale and real-world utility of each category and subcategory in SpatialLab.

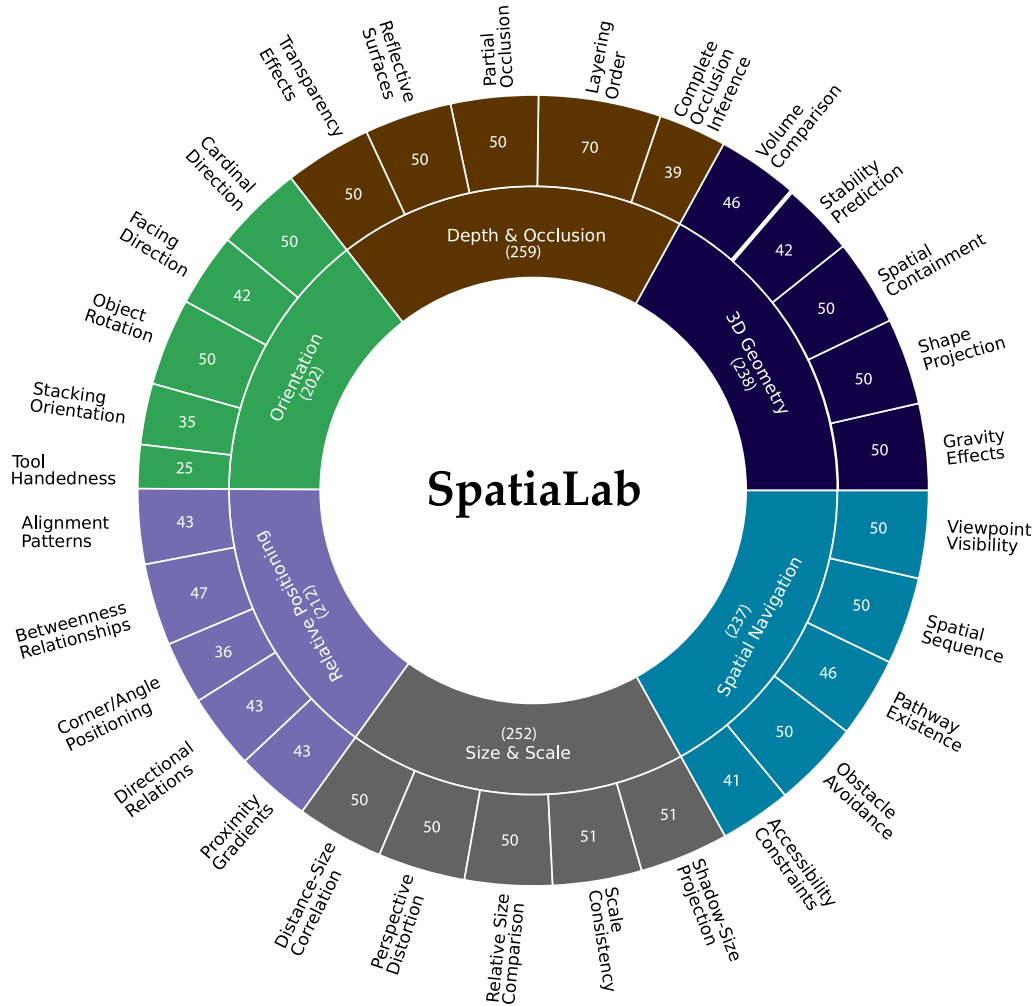


Figure 5: **Spatial-reasoning taxonomy in SPATIALAB.** Each primary category is decomposed into five subcategories, yielding thirty task types.

The following outlines the rationale and real-world utility guiding each task design, per category:

B.1 RELATIVE POSITIONING

Directional Relations. Directional relations specify object locations with respect to a reference object, including left, right, above, and below. Correct interpretation of these relations allows a model to recover spatial layout structure from natural-language queries. For example, identifying the object immediately to the left of an appliance supports scene understanding and robot navigation.

Proximity Gradients. Proximity reasoning requires estimating relative distance from a reference point and determining which objects are nearest or farthest. Questions such as “Which object is nearest to the doorway?” require an ordered distance representation. Models that estimate proximity gradients can support navigation, retrieval, and assistive technologies.

Alignment Patterns. Alignment reasoning evaluates whether objects follow a common orientation, line, or ordering pattern, such as vertical stacking or linear arrangement. Recognizing these configurations supports interpretation of structured environments, including shelves, tables, and warehouses. The same mechanism is also relevant to robotic sorting and automated inspection.

1242 **Betweenness Relationships.** Betweenness requires identifying an object located between two
1243 reference entities. A query asking which object sits between the lamp and the armchair requires
1244 reasoning over spatial continuity and relative ordering. This capability supports navigation, object
1245 retrieval, and situational awareness.

1246 **Corner/Angle Positioning.** Corner and angle relations use bounded regions as spatial anchors.
1247 Questions such as “What’s positioned in the back-right corner?” require localization within a
1248 constrained reference frame. This reasoning is important for robotic mapping, placement, and scene
1249 indexing.

1250

1251 B.2 DEPTH & OCCLUSION

1252

1253 **Layering Order.** Layering order requires distinguishing which objects are visually or physically in
1254 front of others. Identifying the frontmost item on a table is a depth-ordering judgment. This capability
1255 is relevant to AR/VR systems in which correct visual stacking affects interaction fidelity.

1256 **Partial Occlusion.** Partial occlusion occurs when one object hides a visible subset of another object.
1257 Recognizing an item partly hidden behind a coffee cup requires inferring missing structure from
1258 incomplete evidence. This operation improves robustness in cluttered perception settings.

1259 **Complete Occlusion Inference.** Complete occlusion inference requires detecting fully hidden
1260 objects from indirect cues such as shadows, deformation, or contextual evidence. Inferring an object
1261 under a blanket requires the model to use evidence not directly contained in object pixels. This task
1262 approximates human inference under visual uncertainty.

1263 **Transparency Effects.** Transparent media such as glass or water make background objects partially
1264 visible while altering their appearance. Determining what is visible through a fish tank requires
1265 combining occlusion reasoning with transparency perception. This capability is important for
1266 simulation, graphics, and embodied AI.

1267 **Reflective Surfaces.** Reflective-surface reasoning requires interpreting mirrors, polished metals, or
1268 glass reflections as indirect spatial evidence. Identifying what appears in a mirror involves mapping
1269 reflected content into the scene reference frame. This ability is relevant to autonomous driving safety
1270 and robotic situational awareness.

1271

1272 B.3 ORIENTATION

1273

1274 **Cardinal Direction.** Cardinal-direction reasoning maps object or scene orientation to global direc-
1275 tions such as north, south, east, and west. Determining which direction a front door faces requires
1276 integrating global orientation cues. This capability supports navigation and geospatial alignment.

1277 **Object Rotation.** Object-rotation reasoning estimates how far an object has turned around an axis.
1278 Assessing the rotation of a chair supports accurate 3D modeling and object-state estimation. Robots
1279 also require this capability to align objects for manipulation.

1280 **Facing Direction.** Facing-direction reasoning identifies the orientation of an object’s front surface.
1281 For example, determining whether a monitor faces north requires mapping surface orientation to a
1282 global reference frame. This capability is relevant to surveillance, robotics, and scene understanding.

1283 **Stacking Orientation.** Stacking orientation requires determining whether objects are arranged
1284 vertically, horizontally, or in mixed configurations. Correct recognition of these layouts supports
1285 stability assessment in warehouses and homes. It is also necessary for safe robotic stacking.

1286 **Tool Handedness.** Tool-handedness tasks require detecting whether an object such as scissors or a
1287 glove is configured for left- or right-handed use. This judgment depends on fine-grained shape and
1288 orientation cues. It supports assistive robotics in human-centered environments.

1289

1290 B.4 SIZE & SCALE

1291

1292 **Relative Size Comparison.** Relative size comparison requires deciding whether one object is larger,
1293 taller, or smaller than another. Determining whether a vase is taller than a statue requires proportional
1294 reasoning. This operation supports packing, fitting, and object recognition.

1295

Perspective Distortion. Perspective distortion occurs when distance from the camera changes apparent size. Correctly interpreting a distant car that appears smaller than a nearby bicycle requires compensating for depth cues. This capability improves robustness to camera viewpoint variation.

Distance-Size Correlation. Distance-size correlation requires jointly estimating object distance and apparent size. For example, a nearby bicycle may appear larger than a distant car even if its true physical size is smaller. This reasoning supports navigation, AR/VR, and visual grounding.

Scale Consistency. Scale consistency evaluates whether object sizes are plausible in context. A mouse larger than a chair indicates a scale inconsistency that the model should detect. This reasoning supports realism checks in simulated and generated environments.

Shadow-Size Projection. Shadow-size projection uses shadows as indirect evidence about object size and light-source geometry. Estimating object height from shadow length requires geometric reasoning. This task is relevant to graphics, architecture, and physics-based modeling.

B.5 SPATIAL NAVIGATION

Pathway Existence. Pathway-existence reasoning determines whether a traversable route connects two locations. Determining whether one can reach a desk without moving chairs tests navigability under spatial constraints. This capability is foundational for indoor robot planning.

Obstacle Avoidance. Obstacle avoidance requires identifying routes that minimize or prevent collisions. Selecting the clearest path from a door to a window maps directly to robotic and human navigation. It is also relevant to autonomous driving and indoor mobility aids.

Viewpoint Visibility. Viewpoint-visibility reasoning determines which objects are visible or occluded from a specified observer position. For example, identifying what is hidden when sitting on a couch requires simulating perspective. This mechanism supports AR/VR realism and robotic vision.

Spatial Sequence. Spatial-sequence reasoning identifies the order in which objects or regions would be encountered along a path. Walking from a kitchen to a balcony may require passing the dining table first. This task approximates human navigation memory and route understanding.

Accessibility Constraints. Accessibility reasoning identifies objects that cannot be reached because other objects or barriers block access. A cabinet may be unreachable because a table obstructs the approach path. This capability supports assistive AI and ergonomic design.

B.6 3D GEOMETRY

Volume Comparison. Volume comparison requires determining which object occupies more three-dimensional space. Judging whether a box has greater volume than a sphere supports packing and containment tasks. It is relevant to logistics and simulation.

Stability Prediction. Stability prediction estimates whether adding, removing, or moving objects will cause a configuration to topple. Determining whether a book will collapse a pile approximates intuitive physical reasoning. This capability supports safety in robotics and construction.

Shape Projection. Shape-projection tasks require predicting the two-dimensional shape produced by an object under rotation, viewpoint change, or lighting. A sphere projects as a circle, whereas a cube may project as a square or hexagon. This task evaluates geometric abstraction.

Spatial Containment. Spatial containment evaluates whether one object can fit inside another. Determining whether a ball can fit in a box requires size and shape compatibility reasoning. This capability supports packing, design, and intuitive geometry.

Gravity Effects. Gravity-effect reasoning predicts motion or final state under physical forces. For example, a rolling ball may stop under a table because of friction and barriers. This task evaluates physically grounded spatial simulation.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

Annotators were recruited from engineering-graduate populations and were between 24 and 30 years old. Participation was voluntary, and annotators worked independently without external incentives. Before annotation began, we conducted structured training sessions to specify task requirements and standardize interpretation across annotators. Ethical and quality-control procedures were maintained throughout the process. This recruitment and training protocol was intended to produce annotations that are reliable and aligned with the benchmark design.

C.2 ANNOTATOR TRAINING AND PROTOCOL

Before annotation, we implemented a structured training program to increase consistency and reliability across the complete task taxonomy. Annotators were first introduced to the six core categories (*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*) and the corresponding thirty subcategories. For each subcategory, we provided curated positive examples illustrating the relevant spatial relation, such as directional relations, occlusion inference, or stability prediction, together with counter-examples that clarified common failure modes and boundary cases.

Beyond the taxonomy itself, annotators were trained to consider six meta-dimensions of image variability: lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints. The objective was to ensure that questions explicitly used natural visual diversity rather than relying only on easily separable object labels. We demonstrated how these factors interact in real-world scenes, including occlusion under low-contrast lighting and scale consistency under perspective distortion.

Annotators were also trained on the two evaluation modes. In the multiple-choice setting, they learned to construct precise distractors that test fine-grained discrimination rather than superficial answer elimination. In the open-ended setting, they practiced writing prompts that require compositional spatial reasoning. Calibration sessions required annotators to generate both formats for sample images and then iteratively refine the items based on feedback.

To contextualize the task design, we reviewed prior benchmarks, including SpatialBot-Bench, EmbSpatial, and SpatialMM, and analyzed their limitations with the annotators. These benchmarks often emphasize simplified relations, synthetic or staged scenes, and narrow reasoning types, while underrepresenting natural complexity such as shadows, reflections, and gravity-sensitive stability. This comparison motivated the use of richer, more ambiguous, and dynamically layered spatial contexts in SPATIALAB.

Finally, we provided demonstrations of the complete annotation protocol. The demonstrations specified how to select images with sufficient spatial complexity, how to map each question unambiguously to a subcategory, and how to avoid trivial formulations. Annotators were instructed to prioritize items requiring multi-step inference, such as depth layering, relational chains, and cross-view scale reasoning, rather than binary labels. Through this training and iterative feedback process, the final dataset was constructed to balance taxonomic coverage with real-world relevance.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

D.1 MULTIPLE CHOICE EVALUATION

The prompts used for the answer model are shown in Figure 6. We set `temperature` to 0 and `max_tokens` to 10, with all other generation parameters kept at their default values.

D.2 OPEN-ENDED EVALUATION

The prompts used for both the answer model and the judge model are provided in Figure 6. We set `temperature` to 0.7 and `max_tokens` to 1200, with all other generation parameters kept at their default values.



Figure 6: Evaluation prompts used for answer generation and automatic judging.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To evaluate *Gemini-2.5-Flash* as an automatic judge, we selected a balanced validation set of 240 samples spanning all categories. Each sample contained the full evaluation tuple: question, ground-truth answer, and model output. Three human annotators independently assigned binary labels, with 1 denoting a correct response and 0 denoting an incorrect response. We compared the LLM judgments against the human labels using complementary agreement metrics computed both overall and by category. Specifically, we report **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**, defined below.

Cohen’s Kappa. Cohen’s kappa (κ) measures pairwise agreement between two raters while correcting for chance agreement. For binary task, it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, where p_o is the observed agreement and p_e is the expected agreement by chance: $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here, $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities that each rater assigns class c . A value of $\kappa = 1$ indicates perfect agreement, $\kappa = 0$ indicates chance-level agreement, and $\kappa < 0$ indicates systematic disagreement. We computed Cohen’s kappa between the LLM and each annotator individually (κ_{Human1} , κ_{Human2} , κ_{Human3}), as well as between the LLM and the annotator majority vote (κ_{majority}).

Accuracy with Majority Vote. We also measured raw agreement between the LLM and the annotator majority. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} is the LLM’s decision for sample i , and y_i^{maj} is the majority vote among annotators. Unlike Cohen’s kappa, accuracy does not correct for chance agreement; it measures only the proportion of identical labels.

Fleiss’ Kappa. To assess inter-annotator reliability, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa to multiple raters. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, where n is the number of annotators. The observed agreement is $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, and the expected agreement is

$\bar{P}_e = \sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa is then computed as $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Higher values indicate stronger inter-annotator reliability beyond chance.

Table 5: Agreement metrics between Gemini-2.5-Flash, used as an automatic judge, and three human annotators across all categories. The table reports Cohen’s kappa against majority vote and each annotator, raw accuracy against majority vote, and Fleiss’ kappa among annotators; bold values denote the highest agreement within each category.

Category	κ_{Majority}	κ_{Human1}	κ_{Human2}	κ_{Human3}	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Interpretation Table 5 shows that the LLM judge (Gemini-2.5-Flash) achieves strong overall agreement with human annotators. Cohen’s kappa is 0.738 against majority vote and ranges from 0.681 to 0.795 against individual annotators. Raw accuracy against majority vote is 0.880 overall, and Fleiss’ kappa among human annotators is 0.774, indicating substantial inter-annotator consistency. Category-level results show perfect agreement for 3D Geometry, with $\kappa = 1.000$ and accuracy of 1.000. Depth & Occlusion also shows high agreement ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909). Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714) exhibit moderate agreement but remain within acceptable reliability ranges.

These results indicate that Gemini-2.5-Flash approximates human evaluators for binary correctness scoring. Most Cohen’s kappa values lie in the substantial range (0.61–0.80), and some categories reach almost-perfect agreement (above 0.81). Fleiss’ kappa of 0.774 confirms that the human reference labels are themselves reliable. The automatic judge reaches perfect or near-perfect agreement in categories with comparatively well-specified visual evidence, such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933). Moderate agreement in Orientation and Size & Scale is consistent with greater task ambiguity and category-specific subjectivity. Overall, the LLM-as-judge protocol provides a reliable automated scoring mechanism that closely tracks human evaluation.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. Closed-source models (Table 6) produce several of the highest scores in the benchmark. The strongest individual result is obtained by **GPT-4o-mini**, which reaches 85.71% in *Stacking Orientation*. **Gemini-2.0-Flash** performs strongly on occlusion and projection tasks, obtaining 64.0% in *Partial Occlusion* and 64.71% in *Shadow-Size Projection*. **GPT-5-mini** reaches 68.0% in *Tool Handedness* and 55.71% in *Layering Order*. Nevertheless, failures persist; for example, **Claude 3.5 Haiku** reaches only 32.0% in *Relative Size Comparison*. Averaged across several categories, closed-source systems exceed open-source systems by approximately 15–25 percentage points, but within-model variation often exceeds 50 percentage points. Thus, closed-source training pipelines increase the attainable ceiling while failing to produce uniform spatial competence.

Open-source Large Models. Large open-source models (Table 7) show higher peak performance than smaller models but remain highly fragile. **InternVL3.5-72B** obtains 80.56% in *Corner/Angle Positioning*, one of the strongest open-source results. Both **InternVL3.5-72B** and **llama-3.2-90b-vision-instruct** reach 66.67% in *Complete Occlusion Inference*. However, severe failures remain, including **llama-3.2-11b-vision-instruct** scoring only 2.0% in *Object Rotation*. Gains in *Stacking Orientation*, where **llama-3.2-90b-vision-instruct** scores 71.43%, coexist with losses in other categories. The within-model spread frequently exceeds 70 percentage points, indicating brittle specialization

Table 6: Multiple-choice evaluation accuracy on SPATIALAB-MCQ disaggregated by question subcategory for Closed-source Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	42.0	34.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

Table 7: Multiple-choice evaluation accuracy on SPATIALAB-MCQ disaggregated by question subcategory for Open-source Large Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	InternVL3.5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	61.7	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

rather than general capability. Gravity reasoning remains limited at 56.0%. These results indicate that scaling raises selective performance peaks but does not remove structural representational weaknesses.

Open-source Small Models. Small open-source models (Table 8) exhibit sharp subcategory-level contrasts. For example, **qwen-2.5-vl-7b-instruct** reaches 74.29% in *Stacking Orientation*, whereas **InternVL3.5-1B** falls to 12.0% in *Cardinal Direction*. *Pathway Existence* remains difficult, with the strongest models tied at 34.78%. Gravity-related reasoning is also weak, with only 50.0% achieved by **InternVL3.5-4B** and **Gemma-3-4B-it**. Within-model spreads often exceed 60 percentage points, indicating unstable generalization. These patterns suggest that smaller models are strongly cue-dependent: they can perform well when visual templates are salient but fail on tasks requiring deeper spatial abstraction. Overall, small models provide localized competence without robustness across the full spatial taxonomy.

Table 8: Multiple-choice evaluation accuracy on SPATIALAB-MCQ disaggregated by question subcategory for Open-source Small Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

Reasoning Models. Reasoning-enhanced models (Table 9) provide measurable gains in selected tasks but do not achieve consistent coverage. **o4-mini** reaches 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*, outperforming many open baselines. **Gemini-2.0-Flash-Thinking** obtains 62.75% in *Shadow-Size Projection* and 52.17% in *Volume Comparison*, suggesting improved geometric inference. The strongest results in this group come from **Gemini-2.5-Flash-Thinking**, with 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*. However, the same family remains weak in other subcategories, including 34.0% in *Cardinal Direction* and 34.0% in *Shape Projection*. This variance shows that reasoning mechanisms increase performance depth in certain subcategories but do not broaden the coverage of spatial competence. Within-model spreads frequently exceed 35 percentage points, so reasoning augmentation should be interpreted as localized capacity rather than general spatial understanding.

Spatial Reasoning Models. Spatially specialized models (Table 10) show targeted improvements but limited generality. **SpaceQwen2.5-VL-3B-Instruct** obtains 48.0% in *Distance-Size Correlation* and 49.02% in *Scale Consistency*, surpassing many open-source baselines in size-scale reasoning. The same model leads navigation tasks with 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** reach 50.0% in *Stability Prediction*, consistent with domain-targeted reasoning. However, performance drops substantially in other tasks, including 24.0% in *Spatial Containment*. These models rarely exceed 55.0%, even in their strongest subcategories, and subcategory spreads frequently exceed 30 percentage points. Specialization therefore produces narrow gains but does not integrate spatial skills into a robust general framework.

Overall Discussion. Across all categories (Table 8, Table 7, Table 6, Table 10, Table 9), the dominant pattern is fragmented competence. Small open-source models peak at 74.29% in *Stacking Orientation* but collapse to 12.0% in *Cardinal Direction*. Larger models raise ceilings, including 80.56% in *Corner/Angle Positioning*, but can still fail catastrophically with 2.0% in *Object Rotation*. Closed-source systems obtain the strongest scores, including **GPT-4o-mini** at 85.71% in *Stacking Orientation*, yet still drop to 32.0% in *Relative Size Comparison*. Specialized models show strengths in navigation (53.66%) and stability (50.0%) but remain capped below 55% overall. Reasoning models achieve gains such as 71.79% in *Complete Occlusion Inference*, while other subcategories remain at 34.0%. Within-model spreads often exceed 50 percentage points, indicating weak internal consistency. Subcategories requiring physical abstraction, including *Gravity Effects*, *Stability Prediction*, and *Spatial Containment*, consistently remain below 35–40% across model groups. We attribute these failures to insufficient stable encodings of orientation and physics, together with overreliance on surface-level correlations. The broad range from 85.71% to 2.0% indicates that models often match

Table 9: Multiple-choice evaluation accuracy on SPATIALAB-MCQ disaggregated by question subcategory for Reasoning Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Table 10: Multiple-choice evaluation accuracy on SPATIALAB-MCQ disaggregated by question subcategory for Spatial Reasoning Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5-VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

patterns rather than perform robust spatial inference. Scaling, proprietary training, and reasoning augmentation each provide benefits, but none yields comprehensive spatial coverage. Progress

Table 11: Distribution of SPATIALAB-MCQ items by the number of evaluated models (0–5) that answered correctly, disaggregated by category and subcategory. A value of “0 indicates that no model answered the item correctly, whereas “5 indicates that five evaluated models answered correctly.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

will likely require architectural mechanisms that encode geometric grounding and physics-aware constraints.

E.2 MODEL FAILURE ANALYSIS

The distribution in Table 11 reveals structured failure patterns across categories and subcategories. 3D Geometry and Size & Scale have relatively balanced distributions, but both retain many items for which no model answers correctly, indicating that even basic geometric invariances are not reliably internalized. Depth & Occlusion has a heavier mid-range tail (2–4 correct), suggesting partial but inconsistent reasoning over layered scenes. Orientation is broadly weak, especially for Stacking Orientation and Tool Handedness, which indicates poor representation of subtle directional cues and embodied affordances. Relative Positioning is mixed: Proximity Gradients shows moderate success,

whereas Corner/Angle Positioning contains many near-zero correct counts, indicating brittleness in compositional reference-frame reasoning. Spatial Navigation is comparatively stronger in Obstacle Avoidance and Viewpoint Visibility, but Pathway Existence remains error-prone when relational chaining is required. Aggregated over the benchmark, no category exhibits uniformly high accuracy, and the Grand Total is distributed across 0–5 bins. This pattern indicates task-specific success rather than general spatial competence.

These results identify representational and reasoning bottlenecks in current vision–language models. High error rates in Stacking Orientation and Tool Handedness imply limited grounding in physics-based and embodied reasoning, which is consistent with training dominated by static internet imagery rather than interactive or task-driven data. Failures in Corner/Angle Positioning and Pathway Existence indicate insufficient multi-step relational chaining and weak recursive spatial logic. Instability in Transparency Effects and Reflective Surfaces reflects difficulty with non-local visual evidence, including secondary reflections and occluded geometry. In contrast, comparatively stronger performance in Obstacle Avoidance may reflect shortcut learning from frequent co-occurrence patterns of walkable paths rather than genuine planning. Overall, the failures indicate that current training regimes emphasize correlation-driven visual–language alignment while underrepresenting embodied, counterfactual, and physics-based reasoning. Addressing these limitations will require richer training distributions, such as embodied simulation and physics-augmented learning, as well as architectures that explicitly model reference frames and relational constraints.

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). Closed-source models achieve the strongest overall performance in open-ended evaluation, but their subcategory results remain highly uneven. GPT-5-mini is the leading model, obtaining the best scores in **3D Geometry** and **Relative Positioning**, with peaks of 58.14% in Directional Relations and 55.81% in Alignment Patterns. In contrast, Gemini-2.5-Flash and Claude 3.5 Haiku score only 16.28% in Proximity Gradients and 12.0% in Perspective Distortion, respectively. This spread demonstrates that even high-performing models remain fragile when the task requires subtle continuous visual inference. The disparities likely reflect differences in the extent to which model pipelines encode relational and geometric structure. Proximity Gradients is especially difficult because it requires graded analogical reasoning rather than discrete classification. GPT-5-mini’s relative success suggests benefits from scale and proprietary training data, but its remaining failures show that closed-source dominance is comparative rather than absolute.

Open-source Large Models (Table 13). Large open-source models exhibit distributed and isolated strengths. llama-3.2-90b-vision-instruct obtains 60.0% in Depth and Occlusion, while GLM-4.5V-106B-MoE leads Spatial Containment with 38.0%. However, several models fail on Proximity Gradients; qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct each obtain only 9.3%. These results indicate that architectural scale improves selected capabilities but does not guarantee coverage of difficult continuous-reasoning tasks. The variation is consistent with heterogeneous training data: models with stronger exposure to depth and geometry data perform better on occlusion and containment, whereas others generalize poorly. Persistent weakness on Proximity Gradients indicates limited optimization for spatial analogies and distance scaling. Inter-model variance is higher in this group than in the closed-source group, reflecting heterogeneous open-source development pipelines. Overall, large open-source models produce encouraging peaks but require targeted pre-training and fine-tuning to obtain consistent coverage.

Open-source Small Models (Table 14). Small open-source models lag behind larger systems overall, but they show localized strengths. qwen-2.5-vl-3b-instruct obtains 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads Spatial Containment (24.0%) and Stability Prediction (21.43%). These successes coexist with severe failures: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B obtains only 2.0% in Shape Projection. The extreme variability is consistent with limited capacity and restricted training diversity, which constrain the ability to learn robust geometric and relational representations. Subcategories requiring precise orientation or hierarchical reasoning trigger complete breakdowns, because small

Table 12: Open-ended evaluation accuracy on SPATIALAB-OPEN disaggregated by question sub-category for Closed-source Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

Table 13: Open-ended evaluation accuracy on SPATIALAB-OPEN disaggregated by question subcategory for Open-source Large Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	26.09	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	16.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	26.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	28.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

representational errors can invalidate the entire response. Their occasional success on narrow tasks shows that compact models can acquire isolated competencies, but their overall accuracy remains bounded by capacity and data limitations.

Reasoning Models (Table 15). Reasoning-oriented models show substantial internal heterogeneity, with Gemini-2.5-flash-thinking far above its peers. It records the best scores in nearly every sub-category, including 75.0% in Tool Handedness and 57.14% in Complete Occlusion Inference. By contrast, Kimi-VL-A3B-Thinking-2506 scores only 6.0% in Spatial Containment and remains weak across categories. This gap likely arises from differences in how the models integrate multimodal evidence into coherent reasoning trajectories. Gemini-2.5-flash-thinking appears better able to construct and query internal scene representations, enabling stronger inference about object interactions and

Table 14: Open-ended evaluation accuracy on SPATIALAB-OPEN disaggregated by question subcategory for Open-source Small Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

Table 15: Open-ended evaluation accuracy on SPATIALAB-OPEN disaggregated by question subcategory for Reasoning Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

physical properties. Weaker models rely more heavily on surface-level patterns, which are insufficient for compositional or causal spatial reasoning. Although explicit reasoning mechanisms provide clear benefits, failures in Shape Projection and Betweenness Relationships show that current reasoning modules are not universally effective.

Spatial Reasoning Models (Table 16). Spatial reasoning models, despite being explicitly designed for this domain, record the lowest open-ended performance overall. Results are fragmented: several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction.

Table 16: Open-ended evaluation accuracy on SPATIALAB-OPEN disaggregated by question subcategory for Spatial Reasoning Models. Bold and underlined entries indicate the best score within the corresponding model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	<u>16.0</u>
	Shape Projection	<u>8.0</u>	<u>8.0</u>	6.0
	Spatial Containment	8.0	<u>10.0</u>	<u>10.0</u>
	Stability Prediction	<u>14.29</u>	<u>14.29</u>	7.14
	Volume Comparison	19.57	<u>23.91</u>	<u>23.91</u>
Depth and Occlusion	Complete Occlusion Inference	7.69	<u>10.26</u>	7.69
	Layering Order	8.57	<u>11.43</u>	4.29
	Partial Occlusion	10.0	<u>12.0</u>	4.0
	Reflective Surfaces	<u>4.0</u>	<u>4.0</u>	2.0
	Transparency Effects	4.0	<u>8.0</u>	2.0
Orientation	Cardinal Direction	18.0	<u>24.0</u>	8.0
	Facing Direction	<u>23.81</u>	<u>23.81</u>	16.67
	Object Rotation	<u>20.0</u>	<u>20.0</u>	18.0
	Stacking Orientation	8.57	8.57	<u>17.14</u>
	Tool Handedness	0.0	4.0	<u>8.0</u>
Relative Positioning	Alignment Patterns	<u>32.56</u>	<u>32.56</u>	23.26
	Betweenness Relationships	<u>4.26</u>	2.13	0.0
	Corner/Angle Positioning	<u>2.78</u>	<u>2.78</u>	<u>2.78</u>
	Directional Relations	<u>11.63</u>	6.98	<u>11.63</u>
	Proximity Gradients	6.98	6.98	<u>9.3</u>
Size and Scale	Distance-Size Correlation	<u>18.0</u>	<u>18.0</u>	16.0
	Perspective Distortion	16.0	<u>20.0</u>	2.0
	Relative Size Comparison	<u>22.0</u>	20.0	8.0
	Scale Consistency	9.8	11.76	<u>13.73</u>
	Shadow-Size Projection	<u>27.45</u>	<u>27.45</u>	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	<u>9.76</u>
	Obstacle Avoidance	10.0	8.0	<u>14.0</u>
	Pathway Existence	6.52	6.52	<u>8.7</u>
	Spatial Sequence	<u>16.0</u>	12.0	<u>16.0</u>
	Viewpoint Visibility	<u>20.0</u>	16.0	8.0

SpaceQwen2.5-VL-3B-Instruct performs slightly better in Volume Comparison (23.91%) and Scale Consistency (13.73%), but these scores remain low. Betweenness Relationships and Corner/Angle Positioning collapse almost entirely, with scores of 0.0% and 2.78%, respectively. These patterns suggest that narrow specialization without sufficient capacity or training diversity limits representational flexibility. In geometric and positional tasks, small representational errors can produce complete response failure. Thus, specialization alone does not produce robust open-ended spatial reasoning; scale, diverse supervision, and generalizable representations are also required.

Overall Discussion. Across closed-source (Table 12), large open-source (Table 13), small open-source (Table 14), reasoning (Table 15), and spatial reasoning models (Table 16), the dominant pattern is fragmented competence. Closed-source models perform best overall, with GPT-5-mini reaching 58.14% in Directional Relations and 55.81% in Alignment Patterns, yet they still fail on Proximity Gradients. Large open-source models achieve peaks such as 60.0% in Depth and Occlusion, but can drop to 9.3% in Proximity Gradients, showing that scale without targeted spatial supervision is insufficient. Small open-source models show occasional localized strengths, such as 26.0% in Relative Size Comparison, but also catastrophic lows such as 0.0% in Betweenness Relationships. Reasoning models demonstrate that explicit reasoning mechanisms can substantially improve performance, with Gemini-2.5-flash-thinking reaching 75.0% in Tool Handedness, whereas less-integrated reasoning systems such as Kimi-VL-A3B fall to single-digit accuracy. Spatial reasoning models underperform most severely, often scoring below 15% and failing completely on Betweenness Relationships. These results indicate three primary limitations: insufficient representations for continuous and analogical spatial cues, structural constraints associated with model scale and training diversity, and brittle reasoning pipelines that do not generalize across tasks. No evaluated approach provides comprehensive coverage, and within-model gaps exceeding 50 percentage points show that current progress remains piecemeal. Future work should integrate robust geometric encodings, physics-informed reasoning, and cross-task transfer to move from isolated gains toward consistent spatial intelligence.

Table 17: Distribution of SPATIALAB-OPEN items by the number of evaluated models (0–5) that answered correctly, disaggregated by category and subcategory. A value of “0 indicates that no model answered the item correctly, whereas “5 indicates that five evaluated models answered correctly.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

F.2 MODEL FAILURE ANALYSIS

The distribution in Table 17 shows persistent weaknesses, with a pronounced skew toward lower bins (0–2 models correct). 3D Geometry is broadly difficult: Shape Projection has 19 items with zero correct models, and Stability Prediction has 10 such items, indicating unreliable physical and geometric reasoning. Depth and Occlusion is the hardest category overall, with the highest zero-correct count (72), especially in Layering Order (20 zero) and Transparency Effects (16 zero), both of which require nuanced occlusion handling and non-local evidence. Orientation shows similar difficulty; Cardinal Direction has some mid-range success, but Stacking Orientation (11 zero) and Tool Handedness (5 zero, with no perfect-success items) reveal brittle embodied directional reasoning. Relative Positioning is inconsistent, with Betweenness Relationships and Proximity Gradients rarely solved by more than two or three models, while Corner/Angle Positioning remains

low but more evenly distributed. Size and Scale is comparatively stronger, with subcategories such as Scale Consistency containing 6 full-success items, suggesting that some models internalize coarse perspective and scaling cues, although Shadow-Size Projection remains difficult with 17 zero-correct items. Spatial Navigation is split: Spatial Sequence and Obstacle Avoidance show occasional full-model success, but Pathway Existence (16 zero) and Accessibility Constraints (10 zero) remain difficult because they require multi-step relational chaining. Overall, across 356 items, zero-correct cases dominate, while only 74 items are solved by all five models, confirming that consistent spatial competence remains elusive.

These findings indicate that the dominant failure sources are representational and inferential rather than artifacts of dataset design. The high error concentration in Depth and Occlusion reflects reliance on superficial appearance features instead of structured scene representations; for example, models may attend to texture rather than infer hidden layers in Layering Order. Failures in Orientation tasks such as Tool Handedness expose the absence of embodied affordance modeling, which is expected when pretraining data rarely includes explicit annotations of handedness or stacking physics. Weaknesses in Relative Positioning and Spatial Navigation indicate limited ability to chain multi-hop spatial relations; Pathway Existence, for instance, requires integrating local connectivity with global layout, which is not well captured by attention-based co-occurrence priors. Partial success in Size and Scale suggests that models exploit frequent statistical regularities in perspective distortion or scale consistency but fail on less common cues such as shadows. Overall, models succeed when visual cues are salient and correlated with training priors, but fail when the task requires counterfactual, compositional, or embodied inference. Addressing these limitations requires architectures that explicitly model reference frames, occlusion layers, and physical dynamics, together with training data enriched by embodied simulation and counterfactual augmentation.

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

This section provides a detailed analysis of the performance gap between multiple-choice (MCQ) and open-ended evaluation formats across the spatial reasoning categories in SPATIALAB, as reported in Table 18.

G.1 QUANTITATIVE SNAPSHOT

Across 25 models, the average MCQ-to-open-ended performance gap is **23.0%**, with a standard deviation of 5.5%. Dataset-wide subtask means are 24.57 for 3D geometry, 23.45 for depth & occlusion, 23.70 for orientation, 23.11 for relative positioning, 23.42 for size & scale, and 22.89 for spatial navigation, yielding an overall average of 23.01. The overall gap ranges from 11.93 to 31.57, with median ≈ 23.34 . At the model-family level, spatial reasoning models have the largest average gap (27.03), whereas reasoning-oriented models have the smallest (19.11). Proprietary models have lower gaps on average (21.09) than many large open-source models (24.51). The human baseline has an overall gap of 22.64. The largest model-level gaps occur for InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%). The smallest gaps occur for Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36). Correlation analysis indicates that spatial navigation is the strongest predictor of the overall gap (Pearson $r = 0.99$), followed by orientation ($r = 0.83$) and 3D geometry ($r = 0.79$).

G.2 PATTERNS AND OBSERVATIONS

Three principal patterns emerge. First, specialist spatial reasoning models exhibit the largest MCQ→open-ended gaps, suggesting that their training or optimization may emphasize classification-style supervision more than free-form generation. Reasoning-focused models exhibit smaller gaps, consistent with a stabilizing effect from chain-of-thought-style training or decoding on generative outputs. Second, spatial navigation is the strongest contributor to aggregate disparity: models that fail on open-ended navigation reasoning drive the overall gap. Orientation and 3D geometry also contribute substantially, whereas relative positioning and size/scale vary more idiosyncratically across systems. Third, model scale alone does not determine format robustness. Smaller models such as Llama-3.2-11B have among the lowest gaps, while very large models such as InternVL3.5-72B have the highest. Finally, negative or near-zero gaps, including depth & occlusion for Llama-3.2-11B at

Table 18: **Performance gap between MCQ and open-ended accuracy (%) ($\downarrow$) across SPATIALAB spatial categories.** Lower values indicate comparable performance across formats, whereas higher values indicate stronger format sensitivity. Bold and underlined entries denote the highest gap within each model group.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
Open-Source Models (Small)							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
Reasoning Models							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
Spatial Reasoning Models							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

−1.98 and relative position for o4-mini at −3.18, indicate that open-ended evaluation can sometimes be more reliable than MCQ when distractors are misleading.

G.3 HYPOTHESES ON ROOT CAUSES

We hypothesize that the observed MCQ→open-ended gaps in SPATIALAB arise from several interacting factors, each supported by the quantitative observations:

1. **MCQ structural advantage.** The MCQ format constrains the output space, allowing models to exploit surface cues or eliminate distractors without constructing the full spatial relation. Across models, the mean MCQ score is systematically higher than the open-ended score, with an average gap of 23.0% and $\sigma = 5.5\%$. Negative gaps for Llama-3.2-11B in depth & occlusion (−1.98) and o4-mini in relative position (−3.18) show that open-ended generation can occasionally outperform MCQ when distractors are misleading, reinforcing that MCQ can inflate or distort measured performance.

Recommendation: MCQ evaluation should be paired with open-ended tasks and subtask-level diagnostics to estimate model competence more accurately and reduce option-based overestimation.

2. **Specialization bias in spatial reasoning models.** Spatial reasoning models have the largest average gaps (27.03%) despite being optimized for spatial tasks, suggesting that their training may rely heavily on categorical or synthetic selection tasks rather than free-form generation. InternVL3.5-72B, a 72B-parameter spatial reasoning model, has the highest overall gap (31.57%), especially in spatial navigation (36.68%) and orientation (34.44%), indicating dependence on classification-like supervision. Correlations between subtask gaps and the overall gap (SpNav: $r = 0.99$, Orientation: $r = 0.83$) show that these format-dependent failures concentrate in sequential and geometric reasoning tasks.

Recommendation: Fine-tuning or instruction-tuning on open-ended spatial generation tasks can reduce reliance on categorical supervision and improve robustness to generative formats.

3. **Instruction-tuning and stepwise decoding.** Reasoning-oriented models, including Gemini-2-Flash-Thinking and o4-mini, exhibit smaller average gaps of approximately 19.11% and lower subtask variance. Gemini-2-Flash-Thinking has an overall gap of 13.0% and modest subtask gaps, including Difference in 3D Geometry at 13.9% and Difference Relative Positioning at 21.03%. This pattern suggests that chain-of-thought or stepwise decoding helps models produce more stable open-ended responses. Reduced correlation between subtask gap variance and overall gap within this family further suggests that instruction-tuned reasoning mitigates format sensitivity.

Recommendation: Chain-of-thought or stepwise decoding strategies and instruction tuning for open-ended spatial tasks can improve generative consistency and reduce dependence on MCQ structure.

4. **Spatio-navigation stresses sequential grounding.** Spatial navigation gaps are the strongest predictor of the overall MCQ→open-ended disparity (Pearson $r = 0.99$). Models with large spatial-navigation gaps, such as SpaceQwen2.5-VL-3B-Instruct (Spatial Navigation gap = 35.87%), also have large overall gaps (29.78%). This indicates that multi-step referential reasoning is disproportionately impaired in open-ended formats. MCQ options may simplify the reasoning chain, thereby masking weaknesses in sequential grounding and relational computation.

Recommendation: Dedicated open-ended spatio-navigation tasks and targeted multi-step reasoning supervision should be used to measure sequential reasoning and reduce MCQ-driven overestimation.

5. **MCQ distractor artifacts.** Negative or near-zero gaps indicate that distractor design can suppress measured MCQ performance. For example, o4-mini has a negative relative-position gap (−3.18%) while maintaining positive gaps elsewhere, implying that distractor quality can depress MCQ accuracy. Conversely, models such as Qwen-VL2.5-32B-Instruct show very large MCQ→open gaps (overall = 29.85%) across multiple subtasks, indicating that MCQ distractors can also exaggerate apparent competence in selection-based tasks.

Recommendation: MCQ distractors should be audited and refined, and MCQ results should be interpreted together with open-ended evaluation to obtain less biased estimates of spatial reasoning ability.

G.4 PRACTICAL IMPLICATIONS

The results show that MCQ and open-ended scores should be reported jointly. Exclusive reliance on MCQ inflates estimates of practical spatial competence, with average disparities of approximately 23%. Subtask-level reporting is also necessary because spatial navigation disproportionately explains the observed differences. Diagnostics that separate classification-based answer selection from generative spatial reasoning can isolate error modes more precisely. Instruction tuning and decoding strategies that encourage stepwise reasoning are promising for reducing MCQ dependence, but they do not remove the need for open-ended evaluation. Finally, MCQ distractor design should be systematically audited to ensure fair comparisons across formats.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

To improve spatial reasoning performance on the visual question benchmark, we evaluate several complementary intervention strategies and analyze their category-specific effects.

H.1 INHERENT REASONING MECHANISMS

We evaluate built-in reasoning mechanisms by comparing standard models with their reasoning-enabled counterparts (Table 2, Table 3). In MCQ evaluation, reasoning generally increases accuracy, although the effect size is category-dependent. Gemini-2.5-Flash-Thinking improves over its non-reasoning variant by +4.64% overall (52.93% vs. 48.29%), with larger gains in Relative Positioning (+13.1%) and Orientation (+8.6%). o4-mini-medium follows the same pattern and reaches the highest overall MCQ accuracy (53.21%), indicating that explicit reasoning scaffolds are particularly useful in structured, discrete-answer settings.

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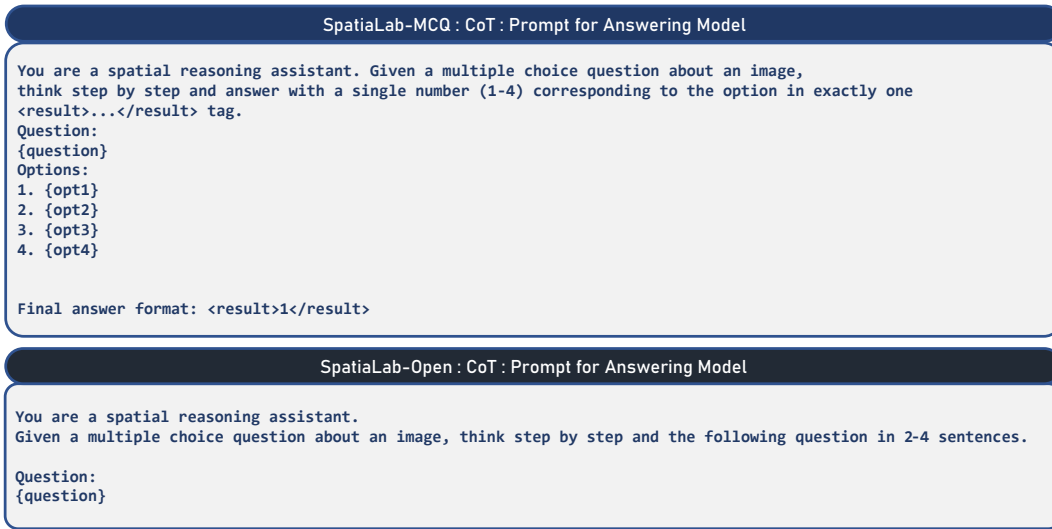


Figure 7: Chain-of-thought (CoT) prompts used to elicit stepwise spatial reasoning.

Open-ended evaluation shows a more heterogeneous effect. Reasoning-equipped models again outperform their baselines in aggregate, but the gains are uneven. Gemini-2.5-Flash-Thinking improves only slightly over Gemini-2.5-Flash (32.77% vs. 30.93%), whereas o4-mini-medium reaches 37.86% and surpasses all non-reasoning models. These results suggest that reasoning mechanisms can reduce instability in free-form generation, especially for categories such as 3D Geometry and Spatial Navigation. However, the effect is not monotonic: Gemini-2.5-Flash-Thinking drops in Size & Scale (21.74% vs. 26.59%), consistent with the finding that generative formats amplify brittleness when perceptual grounding is weak.

Overall, explicit reasoning modules improve structured MCQ performance and can stabilize open-ended responses, but their benefits remain category-specific. The results indicate that reasoning mechanisms strengthen logical consistency and relational chaining while leaving scale sensitivity, occlusion inference, and free-form linguistic grounding only partially addressed.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To test whether structured verbal reasoning improves spatial performance, we evaluate models with Chain-of-Thought (CoT) prompting. For each of the 30 subcategories in SPATIALAB, we sample 5 items with seed 42, producing a balanced 150-item test set that covers all primary spatial phenomena. This design provides uniform category coverage while keeping inference cost tractable. Figure 7 shows the CoT prompts used to provide stepwise reasoning scaffolds. The setup isolates the effect of CoT prompting from changes in model scale or training data.

Table 19: Category-level performance under CoT prompting for InternVL-3-78b and Gemini-2.5-Flash. Values report accuracy (%), and gains denote the difference between CoT and non-CoT prompting.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

The results in Table 19 show that Orientation is the only category in which CoT prompting reliably improves performance; most other categories either stagnate or decline. This pattern suggests that verbal reasoning helps when the task can be reduced to directional alignment, but fails when the



Figure 8: Chain-of-thought (CoT) prompts augmented with a self-reflection and revision stage.

answer depends on perceptual priors such as depth, scale, or relative position. A likely explanation is that the models do not encode sufficiently reliable 3D or geospatial representations. Under CoT prompting, the generated intermediate steps therefore propagate incorrect assumptions instead of correcting them, causing errors to accumulate across the reasoning chain.

This result indicates that SPATIALAB cannot be solved by adding multi-step verbal reasoning to an otherwise unchanged model. Unlike many text-only reasoning tasks, the benchmark exposes architectural limitations in perceptual and spatial grounding. In categories such as depth and size, incorrect perceptual priors cause additional reasoning steps to move the prediction farther from the correct solution. The performance drops, especially for Gemini in MCQ settings, show that reasoning is fragile when it is not anchored in reliable visual-spatial representations.

Thus, the task design probes spatial competence beyond surface-level reasoning. Solving these items requires more than logical chaining; it requires representations and training signals that encode perception, geometry, and spatial cognition. The benchmark therefore exposes a fundamental gap between current reasoning scaffolds and grounded spatial understanding.

H.3 CHAIN-OF-THOUGHT (COT) PROMPTING WITH SELF-REFLECTION

We extend the CoT setup with a self-reflection stage in which the model is instructed to review and revise its initial reasoning before producing the final answer (Figure 8). The evaluation protocol is identical to the standard CoT setting, using 5 balanced samples per subcategory with seed 42. This

design isolates the effect of self-reflection while preserving direct comparability with baseline CoT performance.

Table 20: Category-level comparison of CoT and CoT with Self-Reflection (CoT-SR) for InternVL-3-78b and Gemini-2.5-Flash. Values report accuracy (%), and gains denote the difference between CoT-SR and CoT.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

Adding self-reflection to Chain-of-Thought (CoT) prompting yields mixed and model-dependent outcomes. For InternVL-3-78b, overall accuracy decreases by -2.66% in MCQ and -8.67% in open-ended evaluation, with large drops in depth, relative positioning, and navigation. This indicates that reflection can reinforce errors rather than correct them. Localized gains, such as $+12\%$ in Size & Scale for MCQ, are inconsistent and are offset by large degradations, including -28% in open-ended relative positioning. In contrast, Gemini-2.5-Flash obtains strong MCQ gains ($+20.66\%$), with especially large improvements in 3D Geometry ($+32\%$) and Depth & Occlusion ($+24\%$), while open-ended gains remain negligible ($+0.67\%$). The results reveal a format asymmetry: reflection helps in multiple-choice contexts where unlikely options can be eliminated, but it does not reliably transfer to free-form generation, where uncertainty can propagate into ungrounded rationales. Categories that benefit most from reflection are those requiring geometric consistency, whereas tasks requiring integration across multiple objects, such as relative positioning and navigation, degrade. Thus, reflection operates as a stabilizer for structured answer selection but not as a reliable mechanism for naturalistic spatial reasoning.

This inconsistency is consistent with the plain CoT results: reasoning enhancements succeed only when the model has stable perceptual anchors for the intermediate steps. Self-reflection adds an explicit verification pass, but the pass is constrained by the model’s internal priors. When depth cues, occlusion ordering, or coordinate frames are weakly encoded, the second pass amplifies misconceptions instead of repairing them. This explains why Gemini, which appears to have stronger perceptual grounding, benefits under MCQ conditions, whereas InternVL-3-78b degrades more often. The broader cause is a pretraining mismatch: current VLMs inherit reasoning templates from language modeling but lack the embedded 3D and geospatial representations used in human spatial reasoning. Without these anchors, reflection functions mainly as a linguistic filter that improves reasoning form without improving perceptual grounding. Reflective loops therefore provide limited utility in discrete-answer formats and are unreliable under open-ended demands without external factual verification.

These findings indicate that surface-level reasoning layers cannot compensate for missing spatial cognition. Meaningful progress requires deeper integration of geometric and perceptual structure. Because VLLMs are commonly trained on web data, concepts that are well represented online tend to receive stronger model support, whereas poorly represented spatial and 3D concepts remain underlearned, reinforcing the observed disparities.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We fine-tuned the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) using the parameter-efficient LoRA method (Hu et al., 2021). To fit the model within available GPU memory, we loaded weights in 4-bit precision (`load_in_4bit=True`) and inserted LoRA adapters into attention and feed-forward projection layers. Table 21 reports the LoRA rank, scaling factor, dropout, and target modules. Gradient checkpointing was enabled to reduce memory consumption, and a fixed random seed was used for reproducibility.

For tokenization, we followed the Qwen-2.5 chat template to maintain consistent formatting between training and inference. Each training example was converted into a single text sequence, and dynamic padding was applied with `DataCollatorForSeq2Seq`. Training used an effective batch size of 4 (per-device batch size 1 with gradient accumulation of 4). Trainer and optimization hyperparameters, including learning rate, optimizer, weight decay, and scheduler, are listed in Table 22. The training loop consisted of four passes, each spanning a single epoch, yielding four saved checkpoints.

Overall, this setup allowed us to efficiently fine-tune a large vision-language model on limited GPU resources while maintaining high adaptation capacity.

Table 21: LoRA hyperparameters used for supervised fine-tuning.

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

Table 22: Trainer and optimization hyperparameters used for supervised fine-tuning.

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trends in Figure 4a reveal notable differences between the MCQ and Open-ended evaluation modes over training epochs. We observe that MCQ accuracy begins at a lower baseline of 25.63% but rises consistently, converging near 36.59% by the fourth epoch. This steady improvement suggests that the model benefits strongly from additional supervised fine-tuning, particularly in tasks that constrain answers within a limited candidate space. By contrast, Open-ended evaluation starts higher at 34.40% but declines sharply by the second epoch, dropping to as low as 12.62%. Only in later epochs does it partially recover, reaching 35.48% at the end. This instability points to a sensitivity in open-ended generation that is not present in the MCQ setting, where the structure of multiple-choice questions provides more stable optimization signals. We interpret the fluctuations in open-ended performance as evidence that fine-tuning may overfit to MCQ-like reasoning patterns at the expense of free-form language generalization. The dotted baselines in the plot further highlight how the MCQ mode achieves gains beyond its initial standing, whereas Open-ended remains approximately stagnant. Overall, the figure illustrates that SFT predominantly favors MCQ tasks in terms of stability and consistent accuracy gains.

The tabulated results in Table 4b complement the figure by providing a detailed before-and-after breakdown across six spatial reasoning categories. Across MCQ tasks, we find that performance improved in every category, with gains ranging from a modest 1.59% in Relative Positioning to a substantial 17.11% in Size and Scale. The Grand Total gain for MCQ stands at 10.97%, confirming

that the fine-tuning procedure effectively enhances structured-answer performance. In contrast, Open-ended results are far more mixed, with both positive and negative changes. Categories such as 3D Geometry and Spatial Navigation show small but positive gains (3.50% and 2.82%, respectively), yet Size and Scale exhibits a significant drop of -3.95% . Relative Positioning also decreases by -0.79% , reflecting that fine-tuning can inadvertently harm certain reasoning modes in free-form contexts. The overall Open-ended gain is only 1.07%, which is statistically negligible compared to MCQ improvements. These results confirm that the effectiveness of SFT is highly sensitive to the evaluation paradigm. By isolating the categories, we underscore where training helps, where it fails, and how performance benefits differ between constrained and unconstrained outputs.

A deeper inspection reveals striking disparities between categories and formats. For MCQ, Size and Scale and Spatial Navigation yield the highest improvements, suggesting that models trained with multiple-choice prompts are especially effective at learning tasks requiring scale discrimination and navigational reasoning. However, in Open-ended form, Size and Scale actually declines, showing that the same learned representations do not transfer well to generative contexts. Similarly, Relative Positioning barely improves in MCQ and worsens in Open-ended, highlighting a consistent weakness in relational reasoning regardless of format. Conversely, Orientation shows strong MCQ gains ($+13.33\%$) and small Open-ended improvements, suggesting that some categories generalize more robustly. We hypothesize that the divergence between the two formats may stem from the structural scaffolding MCQs provide, which reduces the need for the model to generate precise linguistic formulations. Open-ended tasks, by contrast, require simultaneous mastery of reasoning and natural language fluency, which amplifies weaknesses. The trends indicate that while fine-tuning strongly enhances pattern recognition in discrete-choice settings, it introduces volatility and even degradation when free expression is required. This duality makes clear that training interventions optimized for MCQs may not generalize seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, we show that SFT as currently applied provides strong benefits for structured tasks but limited or even negative transfer to open-ended reasoning. This limitation is not only an artifact of fine-tuning, but also symptomatic of how contemporary VLMs are pretrained. Existing pretraining pipelines heavily emphasize broad image-text alignment, object recognition, and captioning, yet they provide little explicit supervision on compositional spatial structures, occlusion layers, or geometric transformations. As a result, models learn to excel at surface-level associations while failing to acquire robust representations of depth, orientation, or scale. The instability we observe in open-ended spatial reasoning suggests that the model’s learned priors are fragile: without multiple-choice scaffolding, they revert to shallow pattern-matching instead of grounded inference. Categories such as Size and Scale and Relative Positioning, where declines are most pronounced, point to missing inductive biases in pretraining, biases that humans develop through embodied interaction with 3D environments. Another contributing factor is the reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions, preventing models from generalizing to real-world clutter, noise, and complexity. These findings underscore the need to rethink both pretraining and fine-tuning for spatial reasoning, incorporating datasets and objectives that reflect naturalistic visual-spatial challenges. Ultimately, we argue that without targeted reforms, current VLMs risk reinforcing an algorithmic straightjacket: they appear competent under constrained evaluations yet remain brittle and unaligned with the embodied, flexible reasoning that humans bring to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To investigate the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) using the Qwen-2.5-VL-3B-Instruct model. The training trajectory, summarized in Table 23, reveals a distinct divergence in performance dynamics between Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. As illustrated in Table 23 and Figure 9, performance on SPATIALAB-MCQ exhibits a monotonic improvement, rising from a baseline of 25.63% to an average of 35.74% at Epoch 4. This indicates that the model successfully adapts to the discriminative nature of the task, learning to map visual features to constrained output options with increasing reliability.

In sharp contrast, SPATIALAB-OPEN displays a pronounced *U-shaped* trajectory across all seeds. Rather than steady improvement, the model suffers an immediate and significant performance collapse in the early epochs (dropping from 34.40% to an average of 19.26% at Epoch 1). While performance recovers in later epochs (converging to $\approx 34.82\%$ at Epoch 4), the final gain over the zero-shot baseline is negligible ($+0.42\%$).

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). While MCQ performance improves steadily, Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, indicating representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

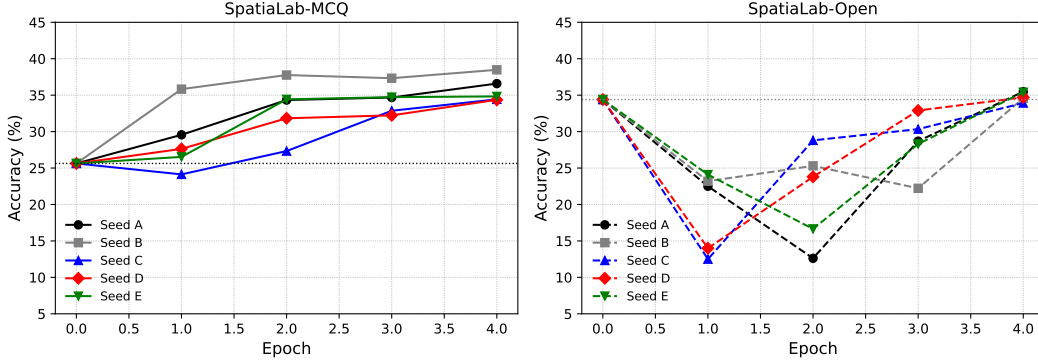


Figure 9: Training dynamics under supervised fine-tuning, comparing MCQ and open-ended behavior across epochs.

Representational Brittleness and Alignment Tax. We attribute this behavior to the fragility of spatial representations in current VLMs under fine-tuning. The initial collapse in open-ended accuracy suggests *catastrophic forgetting* of linguistic priors required for fluent and grounded generation. As the model optimizes for the instruction-following format of the training data, it overfits to the structural logic of the benchmark, which benefits MCQ selection but weakens the flexible generative grounding required for open-ended descriptions.

The subsequent recovery phase indicates that the model partially re-learns to express spatial concepts within the new distribution, but it does not substantially exceed the initial baseline. This dynamic indicates that SFT is effective for teaching models to *select* correct spatial answers, while providing limited improvement to the underlying *generative spatial representation* required for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To assess whether representations learned from SPATIALAB generalize beyond the benchmark format, we evaluate the fine-tuned model on three external spatial reasoning datasets: OMNISPATIAL,

SPACE (Multimodal), and MIND THE GAP. We use Qwen2.5-VL-3B-Instruct, fine-tuned on SPATIALAB with the SFT protocol described in Appendix H.4.1, and evaluate it on the external benchmarks in a zero-shot setting.

Table 24: Transfer learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains indicate that the model learns generalizable spatial representations rather than merely overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As shown in Table 24, fine-tuning on SPATIALAB produces consistent improvements across all evaluated domains, ranging from +5.24% to +11.70%. The largest gain occurs on MIND THE GAP (+11.70%), suggesting that the benchmark emphasis on occlusion and spatial continuity transfers to the reasoning gaps targeted by that dataset. Improvements on OMNISpatial (+7.05%) and SPACE (+5.24%) further indicate that the geometric and relational concepts in the taxonomy align with broader spatial-intelligence evaluations. These transfer results suggest that SPATIALAB does not merely induce overfitting to a single question template. Instead, the diversity of the 30 subcategories encourages acquisition of spatial representations that transfer to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

We developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning on images within the SPATIALAB evaluation. The system decomposes visual reasoning into specialized sub-tasks handled by dedicated agents for objects, attributes, relations, symmetries, and transformations. This modular design supports interpretability and multi-step reasoning by propagating low-level perceptual cues into higher-order relational and transformational inference. The architecture, agent implementations, and coordinated interactions are described below, and the full prompts are provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—including geometric shapes, natural elements, and artificial icons—alongside attributes such as size, orientation, color, and shading. Relative positions like leftmost, center, and top are encoded explicitly, and the prompt emphasizes exhaustive detail without summarization. Images are converted to data URIs and sent with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list using regex-based extraction. This structured description serves as the foundation for downstream agents, providing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach ensures that even subtle visual features, such as partial occlusions or minor shape variations, are retained for later relational and transformational reasoning.

Object Segmentation Agent. Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, differentiating between visually similar instances. Each object receives a stable identifier (e.g., Obj1, Obj2) to maintain consistent references across subsequent reasoning steps. The agent uses carefully formatted prompts and calls the same Gemini-2.5 model to structure the description, while post-processing splits lines by object ID to enforce consistency. This structured representation reduces ambiguity in downstream attribute and relation extraction, enabling precise mapping of complex scenes. By explicitly isolating objects, the agent also facilitates detection of overlaps, repetitions, and hierarchical groupings that may not be evident in raw descriptions.

Attribute Extraction Agent. Attribute Extraction agent translates each object into a structured JSON capturing core properties such as shape, size, color, shading, orientation, and positional

descriptors. Prompts enforce strict JSON output, and responses are parsed using `json.loads()`, with fallback defaults to ensure robustness against model inconsistencies. This structured encoding allows quantitative and qualitative comparisons of objects, supporting later stages of spatial reasoning. By preserving both absolute and relative measures, the agent maintains fidelity to the original perceptual content, while also normalizing descriptors for computational consistency. The JSON structure enables programmatic downstream operations, such as calculating symmetries or detecting transformations.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA,Relation,ObjectB), capturing directional relations (e.g., left_of, above, inside) and higher-order alignments like rows, grids, or repetition patterns. Consistent object IDs are enforced to prevent relational ambiguity, and parsing splits output lines starting with parentheses, followed by automated consistency checks. This step transforms static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure allows reasoning over both pairwise interactions and more complex scene configurations, which is critical for tasks like symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. Grouping and Symmetry agent abstracts higher-order patterns from object lists and spatial relations, identifying clusters, rows, grids, and symmetries—including vertical, horizontal, rotational, and translational. Outputs are requested in structured JSON with keys `groups` and `symmetries`, with empty structures as fallbacks to maintain downstream compatibility. This agent captures emergent scene regularities that are not apparent at the object or pairwise relation level, providing critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent supports both perceptual inference and anticipatory reasoning for dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes including translations, scaling, rotations, color/shading shifts, and appearance or disappearance events. Structured logs are extracted line-by-line using regex patterns that capture directional arrows, producing a temporal representation of the scene. This encoding allows the system to track not just static spatial configurations but also dynamic evolution, supporting reasoning about motion, causality, and object interactions over time. The agent ensures that transformations are linked to consistent object IDs, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates all outputs into a unified JSON format containing `objects`, `relations`, `groups`, `symmetries`, and `transformations`. Prompts enforce consistent object IDs and structured formatting, while JSON parsing failures revert to empty dictionaries to ensure pipeline robustness. By centralizing all perceptual, relational, and temporal information, this agent creates a comprehensive scene representation that is both human-interpretable and machine-readable. The resulting unified structure serves as a versatile foundation for downstream tasks, including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation feeds into the reasoning agent, which answers multiple-choice or pattern recognition questions. Inputs include objects, attributes, relations, groups, symmetries, and transformations. The model considers geometric patterns, relational alignment, and dynamic transformations to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

The implementation uses Gemini-2.5-Flash with low temperature (0.1) to reduce output stochasticity and supports multiple refinement iterations. API calls are routed through OpenRouter and accept both text and image inputs. Model responses are parsed with regex and JSON-based extraction to obtain structured outputs. All interactions are logged in JSON format for reproducibility, and the final output is a unified JSON object per image containing objects, relations, symmetries, and transformations. The decomposition constrains each agent to a specific sub-task, improving interpretability and robustness of the structured reasoning pipeline.

H.5.3 EVALUATION AND ANALYSIS

The evaluation protocol matches the standard CoT and CoT-SR setting: we use 5 balanced samples per subcategory with seed 42, for a total of 150 samples.

Table 25: Agentic spatial-reasoning performance for Gemini-2.5-Flash with and without the multi-agent pipeline.

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67 %	57.33 %	+2.66 %	35.83 %	31.33 %	-4.50 %

The evaluation of Gemini-2.5-Flash with agents identifies both benefits and limitations of agentic reasoning for perceptual-geometric tasks. Orientation is the only category with consistent and substantial gains, increasing by +8.00% in MCQ and +36.00% in open evaluation. This pattern suggests that agent-based multi-step reasoning is effective when a task decomposes into sequential alignment of directional cues, where structured deliberation has an advantage over single-step output generation.

Most other categories exhibit neutral or negative gains, indicating that agentic reasoning does not universally improve performance. Depth & Occlusion decreases by -24.00% in open evaluation, and Spatial Navigation decreases by -12.00%. These tasks require stronger perceptual grounding in three-dimensional structure; when such priors are weak, intermediate agent steps reinforce erroneous assumptions rather than correcting them. Relative Positioning remains unchanged in MCQ and decreases by -14.00% in open evaluation, suggesting that agent loops do not add robustness when the underlying relational representation is unstable.

Some categories show marginal gains, such as Size & Scale (+4.00% in MCQ), but these gains are not stable and do not transfer to open evaluation. In 3D Geometry, the agent pipeline decreases performance by -4.00% in MCQ and -12.00% in open evaluation. These results indicate that the benchmark cannot be solved by layering multi-step agentic reasoning on top of existing model outputs. It requires tighter integration between perceptual representations, geometric structure, and spatial inference.

Overall, the mixed results emphasize that agentic workflows alone are insufficient for advancing performance in this domain. While orientation tasks benefit from structured reasoning, most categories expose the models' lack of inherent awareness of perceptual concepts. This makes errors propagate through multi-step reasoning, amplifying performance drops. Consequently, our benchmark serves as a strong stress test, highlighting that future progress will require embedding genuine perceptual and geospatial priors rather than relying solely on agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section provides a qualitative analysis of model failures observed in the visual reasoning benchmark. We define each error class, provide diagnostic root-cause analysis, identify recurring patterns, and specify verification protocols. The objective is to make the failure modes interpretable and actionable for future architecture and evaluation design.

I.1 OVERVIEW

Across the sample set, models show strong coarse perception: they often recognize object categories, colors, and simple binary relations. However, performance degrades systematically when multiple cues must be integrated. The failures cluster into a small set of interpretable classes rather than appearing as random noise: spatial mislocalization, perspective and scale errors, occlusion and ordering errors, attribute confusion, and ungrounded open-ended rationalization. These classes recur across architectures and prompting strategies, suggesting that they arise from common inductive biases in current vision–language pipelines rather than isolated training artifacts. The same failure type often appears in both proprietary systems and small open-source baselines, indicating shared structural weaknesses across model families.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** When queries require distinguishing one object from a set of visually similar candidates, models frequently misidentify the target. For example, if asked to point out “the leftmost red cube among three stacked cubes,” the model may incorrectly output the middle cube, influenced by lighting highlights or central positioning. This problem becomes particularly acute in cluttered environments such as kitchen counters or crowded table scenes, where multiple overlapping objects share the same color and shape. In such cases, the model’s errors are not random: it tends to privilege objects that appear most salient due to size, brightness, or centrality, even when these cues contradict the explicit spatial instructions in the query. These consistent patterns highlight that the failures are systematic, not accidental.

★ **Analysis.** The recurring mistakes suggest that models lack a robust object-centric representation, or what cognitive science would call an “index” that persists across reasoning steps. Instead of maintaining a stable pointer to the correct candidate throughout the reasoning chain, models appear to drift, often reassigning attention to whichever object seems most visually striking at intermediate layers. This leads to reasoning errors such as starting with the leftmost cube but ending the chain with the brightest or largest cube. The inability to anchor spatial reference consistently makes tasks involving relative comparisons, like “the ball behind the second chair”, particularly fragile. In essence, the model operates more like a salience detector than a referential reasoner.

⊗ **Root cause.** At the architectural level, feature pooling operations, whether via attention or convolution, tend to mix contextual information from neighboring regions. While this is beneficial for global context, it erases the fine-grained, object-specific identity needed to distinguish between similar candidates. Once pooled, features of two adjacent red cubes may be nearly indistinguishable in the latent space. Without an explicit mechanism, such as slot-based representations or pointer tokens, the network collapses distinct entities into blended embeddings. This structural limitation prevents reliable referent tracking across reasoning steps and explains why models consistently confuse dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks involving metric reasoning about relative size, fit, or scale are inconsistently handled. On simple comparisons like “is a car larger than a bicycle,” models typically succeed because such relationships align with strong statistical priors. However, when faced with near-threshold decisions such as “will the small sphere fit inside the slightly larger cylinder,” errors become frequent and unpredictable. A common pattern is reliance on canonical object sizes: for example, predicting that a door must be taller than a truck even in a rendering where the truck is scaled down. Another failure occurs in unusual visualizations, such as toy miniatures or architectural models, where familiar proportions are deliberately inverted. In these cases, models cling to their priors and ignore the actual geometric evidence visible in the image.

★ **Analysis.** These behaviors indicate that models rely heavily on distributional statistics of object categories rather than performing true geometric inference. Instead of reasoning about projection geometry, relative bounding box sizes, or depth cues, they substitute their world knowledge of typical scales. This shortcut works well in everyday contexts but fails catastrophically in edge cases that deviate from the training distribution. For instance, when an unusually small airplane is rendered next to a person, the model still predicts the airplane as larger. Such errors highlight the lack of grounding

in pixel-level metric cues and the absence of consistent use of perspective or vanishing points.

⊗ **Root cause.** Current architectures do not include explicit geometric reasoning modules that preserve projection-aware relationships. Supervision for scale and fit tasks is limited, so models learn correlations between object category labels and size but not the underlying geometry. Furthermore, training objectives prioritize overall accuracy rather than fine-grained metric consistency, meaning that scale-sensitive errors are under-penalized. Without inductive biases for 3D reasoning or targeted supervision, the model defaults to memorized priors about canonical sizes rather than interpreting the actual perspective geometry in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Determining which object is in front, behind, or partially hidden remains a core weakness. Models often misreport the foreground object when two animals overlap, or when semi-transparent barriers like glass or fences are involved. For instance, in an image where a cat sits partially behind a curtain, models may incorrectly describe the curtain as being occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, are particularly error-prone: they are frequently omitted altogether, leading to wrong answers in questions like “what object lies behind the fence.” These mistakes occur consistently, especially when occlusion cues are subtle or when the depth difference is small.

★ **Analysis.** Such failures indicate that models treat depth and occlusion as weak, secondary signals. While the pixel data contains clear occlusion cues (e.g., edge boundaries, T-junctions), the intermediate features learned by the model do not retain them with sufficient resolution. Aggregation layers blur boundaries and downweight thin structures, effectively removing the very cues necessary to resolve ordering. This is why models succeed in gross depth distinctions (foreground vs. background) but collapse in fine-grained cases where partial occlusion must be inferred. In practice, they guess rather than infer, with systematic biases toward treating the visually larger or brighter object as foreground.

⊗ **Root cause.** The architecture lacks explicit mechanisms for multi-scale feature retention and has no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are easily lost when features are pooled across large receptive fields. Furthermore, training data often underrepresents subtle occlusion cases, especially those involving transparency or fine boundaries. As a result, the learned representations flatten layered scenes into 2D appearance maps, preventing robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models sometimes conflate perceptual attributes (such as color, texture, or material) with functional or semantic roles. A classic example is misjudging whether a crosswalk is safe: models may base the answer solely on the presence of painted stripes while ignoring the state of the traffic light or approaching vehicles. Similarly, a surface with a metallic texture may be misclassified as “steel” even if it is painted plastic. In tool-use contexts, the presence of handles or familiar shapes may lead the model to infer functionality that the object does not possess. These confusions are frequent in tasks requiring fine-grained visual grounding of attributes.

★ **Analysis.** These errors reveal that language priors strongly bias the model’s predictions when visual input is ambiguous. The decoder favors plausible narratives based on co-occurrence in the training data: “crosswalks are safe when stripes are visible,” “shiny textures correspond to metal,” and so on. This shortcut allows models to produce semantically coherent answers even when they do not inspect the relevant visual evidence. The result is superficially correct reasoning in familiar contexts but systematic errors in atypical or adversarial scenarios. In practice, this manifests as overfitting to textual stereotypes rather than grounding in perceptual details.

⊗ **Root cause.** The underlying training objective, maximizing likelihood of fluent and plausible text, does not enforce visual grounding. As long as the generated answer is semantically reasonable, the loss is minimized, even if the reasoning is incorrect. Consequently, the model is incentivized to substitute common-sense priors for genuine perceptual inference. Without auxiliary losses or verification steps to tie textual claims to image regions, models will continue to swap perceptual and semantic attributes.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** For explanatory or why-style questions, models often generate coherent but unsupported narratives. For example, when asked “why is the object leaning,” a model might answer “because the surface is slippery” even though slipperiness cannot be visually inferred. In another case, a tilted shelf might be explained as “due to an earthquake,” a claim entirely outside the evidence in the image. These errors highlight a tendency to fill gaps with imaginative but ungrounded rationales, producing fluent explanations that mislead users about the true evidence.

★ **Analysis.** The issue stems from a decoupling between text generation and visual grounding. The model generates high-quality language that reads convincingly, but there is no mechanism ensuring that these rationales are tied to actual visual features. Confidence calibration compounds the problem: models often assign high certainty to fabricated explanations, giving the false impression of reliability. As a result, open-ended outputs may look authoritative while being factually baseless. This mode of failure is particularly concerning in safety-critical domains where plausible but false rationales could misguide decision-making.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence but not factual grounding. Decoders are optimized to produce the most probable continuation of text given the prompt, without constraints linking the explanation back to specific image evidence. The lack of an explicit verification layer or grounding requirement means that models default to linguistic plausibility over perceptual truth. Without additional architectural or training constraints, open-ended rationalizations will continue to prioritize surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

When comparing failures across different models, several consistent patterns appear, revealing insights into the underlying mechanisms of errors. One clear trend is that disagreements between models are systematic rather than random: some models consistently rely on geometric cues such as relative size, depth ordering, and occlusion relationships, while others fall back on linguistic priors or common-sense assumptions derived from training text. This leads to predictable divergences, for example when two models give different answers about which object is “in front” despite seeing the same scene. Errors also spike when multiple cues must be integrated simultaneously. Tasks that require reasoning about occlusion, relative scale, and perspective together are particularly challenging; for instance, understanding whether a larger object partially occluding a smaller one is in front or behind often triggers consistent mispredictions. Calibration is another noticeable issue: high-confidence predictions frequently correspond to incorrect answers, especially in open-ended explanations or complex spatial queries. These patterns indicate that errors arise from how models balance perceptual grounding and linguistic plausibility. Different architectures exhibit distinct tendencies, some over-rely on visual geometry, others overfit to language correlations, so failures are not just isolated mistakes but reflect deeper trade-offs in reasoning strategies. This also implies that any attempt to improve performance should account for both sources of bias simultaneously rather than treating errors as independent events.

I.4 MITIGATION STRATEGIES

Given these error patterns, several targeted strategies can be pursued to reduce failures without requiring complete architectural redesign. For instance, geometry-aware supervision can be incorporated by adding auxiliary tasks like depth prediction, occlusion ordering, or overlap estimation. This encourages models to respect spatial constraints and improves consistency across predictions. Multi-scale feature retention is another approach: by using edge-aware pooling or attention mechanisms, models can maintain information from thin or partially occluded structures that would otherwise vanish in standard downsampling. Coordinate anchoring can help resolve confusion between viewer-centric and object-centric frames by explicitly representing reference points for relative positioning. Counterfactual augmentation can also be applied, where distractor objects are systematically swapped or perturbed, forcing the model to rely on the true referent cues rather than heuristics. Lightweight verification modules can be added on top of existing outputs, cross-checking claims against visual or geometric constraints to filter implausible predictions. Each of these strategies can be implemented incrementally, providing clear paths to reduce systematic failures while keeping the core model largely unchanged.

In addition to these architectural and training augmentations, supervised fine-tuning (SFT) offers a practical pathway to instill more reliable spatial reasoning. By curating datasets that explicitly highlight failure cases, such as near-threshold scale comparisons, occlusion ambiguities, or reference resolution traps, SFT can directly expose models to the edge scenarios where they otherwise default to heuristics. Carefully designed instruction-style fine-tuning further encourages step-by-step reasoning, prompting the model to verbalize intermediate spatial relations rather than skipping to a guess. This not only strengthens grounding but also improves interpretability, as errors become traceable to specific reasoning steps. Moreover, SFT can be paired with counterfactual or adversarial augmentations, ensuring that the model does not simply memorize but instead learns invariances to distractors and context shifts. Importantly, these interventions require no wholesale changes to the architecture: they adapt existing models by reshaping their inductive biases through supervision. In this sense, SFT serves as a lightweight yet powerful complement to geometry-aware and verification-based strategies.

In this work, we additionally performed SFT using 40% of the training data and observed clear performance gains, particularly on multiple-choice questions. This suggests that targeted supervision helps models better internalize spatial cues and resolve distractor confusions when answer options are well-defined. However, improvements on open-ended responses were more limited. The model continues to produce fluent but weakly grounded explanations, highlighting that SFT alone cannot fully address rationalization without verification. These findings reinforce the need to combine SFT with geometry-aware supervision, counterfactual augmentations, and lightweight verifiers to achieve balanced progress across both structured and free-form tasks.

Our analysis shows that failures are not arbitrary but structured and interpretable, emerging from recurring patterns rather than random noise. This indicates that VLM errors follow systematic trajectories tied to the inductive biases of their architectures and training regimes. For example, geometric reasoning often collapses when occlusion cues are weak, and in such cases language priors take over, producing answers that sound plausible but lack visual grounding. Similarly, scale inference drifts when canonical object size priors dominate over pixel-level metric cues, leading to consistent misjudgments in near-threshold scenarios. These findings highlight that the gap is not simply one of capacity but of representation: current models lack explicit mechanisms to anchor visual features to grounded geometric relations.

By adopting the proposed verification protocols and targeted mitigation strategies, researchers can address these weaknesses directly. Geometry-aware supervision and counterfactual augmentation would help models resist shortcuts, while multi-scale feature retention ensures thin structures and partial occlusions are not discarded. SFT, when carefully applied, can further strengthen structured reasoning by aligning model predictions with task constraints, though it must be combined with grounding objectives to avoid overfitting to discrete-choice formats. Importantly, verification modules provide a lightweight way to enforce consistency between claimed relations and image evidence, counteracting the tendency to rationalize without checking. Taken together, these steps chart a practical pathway toward systems that reason more like humans: robust to clutter, interpretable in their failures, and capable of explaining not only what they predict but why.

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Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
 - Obj1: Small shaded circle (leftmost)
 - Obj2: Large unshaded square (center)
 - Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
 - JSON array, e.g.:

```
[  
  {  
    "ID": "Obj1",  
    "Shape": "Circle",  
    "Size": "Small",  
    "Shading": "Yes",  
    "Rotation": "0°",  
    "Position": "Leftmost"  
  },  
  {  
    "ID": "Obj2",  
    "Shape": "Square",  
    "Size": "Large",  
    "Shading": "No",  
    "Rotation": "45°",  
    "Position": "Center"  
  }  
]
```

Figure 10: Prompt specification for SPATIOXOLVER (Part 1).

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Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompt specification for SPATIOXOLVER (Part 2).

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Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{  
  "objects": [...],  
  "relations": [...],  
  "groups": [...],  
  "symmetries": [...],  
  "transformations": [...]}  
}
```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompt specification for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

We present representative benchmark samples for all 30 subcategories in SPATIALAB. The examples instantiate the main visual phenomena evaluated by the benchmark, including 3D geometry, occlusion, orientation, relational positioning, size and scale, and navigation. Collectively, they specify the reasoning operations that models must perform to succeed across the full taxonomy.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Representative benchmark sample and evaluation outputs for 3D Geometry: Gravity Effects.

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3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- **Gemini-2.5-Flash:** The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- **InternVL3.5-72B:** The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Representative benchmark sample and evaluation outputs for 3D Geometry: Shape Projection.

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3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Representative benchmark sample and evaluation outputs for 3D Geometry: Spatial Containment.

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3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Representative benchmark sample and evaluation outputs for 3D Geometry: Volume Comparison.

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3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- Gemini-2.5-Flash: The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- InternVL3.5-72B: From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Representative benchmark sample and evaluation outputs for 3D Geometry: Shape Projection.

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Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Representative benchmark sample and evaluation outputs for Spatial Navigation: Accessibility Constraints.

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Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

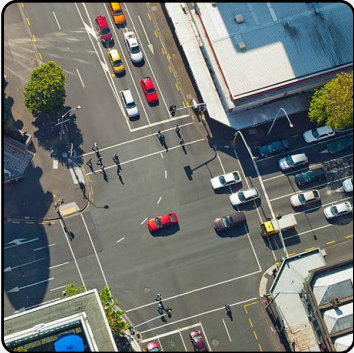
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Representative benchmark sample and evaluation outputs for Spatial Navigation: Obstacle Avoidance.

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Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

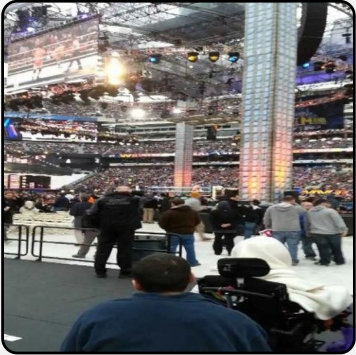
Open-ended Answers:

- **GPT-5-mini:** The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- **Gemini-2.5-Flash:** The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- **InternVL3.5-72B:** The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Representative benchmark sample and evaluation outputs for Spatial Navigation: Pathway Existence.

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Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Representative benchmark sample and evaluation outputs for Spatial Navigation: Viewpoint Visibility.

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Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Representative benchmark sample and evaluation outputs for Spatial Navigation: Spatial Sequence.

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Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Representative benchmark sample and evaluation outputs for Size and Scale: Scale Consistency.

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Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Representative benchmark sample and evaluation outputs for Size and Scale: Shadow-Size Projection.

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Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Representative benchmark sample and evaluation outputs for Size and Scale: Perspective Distortion.

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Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- Gemini-2.5-Flash: The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- InternVL3.5-72B: The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- Kimi-VL-A3B-Thinking-2506: Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position. The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Representative benchmark sample and evaluation outputs for Size and Scale: Relative Size Comparison.

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Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Representative benchmark sample and evaluation outputs for Size and Scale: Relative Size Comparison.

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Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- **Gemini-2.5-Flash:** The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- **InternVL3.5-72B:** The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- **Kimi-VL-A3B-Thinking-2506:** The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Representative benchmark sample and evaluation outputs for Depth and Occlusion: Complete Occlusion Inference.

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Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Representative benchmark sample and evaluation outputs for Depth and Occlusion: Layering Order.

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Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Representative benchmark sample and evaluation outputs for Depth and Occlusion: Partial Occlusion.

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Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Representative benchmark sample and evaluation outputs for Depth and Occlusion: Reflective Surfaces.

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Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

- 7
- 6
- 5
- 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Representative benchmark sample and evaluation outputs for Depth and Occlusion: Transparency Effects.

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Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Representative benchmark sample and evaluation outputs for Orientation: Cardinal Direction.

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Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

1. Up
2. Down
3. Away
4. Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- **GPT-5-mini:** If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- **Gemini-2.5-Flash:** The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- **InternVL3.5-72B:** If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- **Kimi-VL-A3B-Thinking-2506:** The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Representative benchmark sample and evaluation outputs for Orientation: Facing Direction.

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Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

1. 90
2. 45
3. 180
4. 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Representative benchmark sample and evaluation outputs for Orientation: Object Rotation.

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Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Representative benchmark sample and evaluation outputs for Orientation: Stacking Orientation.

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Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Representative benchmark sample and evaluation outputs for Orientation: Tool Handedness.

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Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Representative benchmark sample and evaluation outputs for Relative Positioning: Alignment Patterns.

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Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- **Gemini-2.5-Flash:** The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- **InternVL3.5-72B:** The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- **Kimi-VL-A3B-Thinking-2506:** The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Representative benchmark sample and evaluation outputs for Relative Positioning: Betweenness Relationships.

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Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Representative benchmark sample and evaluation outputs for Relative Positioning: Corner/Angle Positioning.

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Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Saffari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Representative benchmark sample and evaluation outputs for Relative Positioning: Directional Relations.

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Relative Positioning : Proximity Gradients



Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- Gemini-2.5-Flash: "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- InternVL3.5-72B: The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- Kimi-VL-A3B-Thinking-2506: Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Representative benchmark sample and evaluation outputs for Relative Positioning: Proximity Gradients.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

To verify that benchmark conclusions are not artifacts of random initialization or dataset noise, we evaluate model robustness across repeated runs and dataset stability under resampling. These analyses are important for small- to medium-scale benchmarks, where minor fluctuations can otherwise produce misleading fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010).

Let N denote the number of benchmark items (questions), and let $R = 3$ denote the number of independent runs for the same model. Each run uses a distinct random seed while all other hyperparameters remain fixed.

K.1 MODEL ROBUSTNESS

To assess model-level evaluation robustness, we conduct the statistical analyses described below.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is:

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

and its corresponding standard deviation is:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy per run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and its standard deviation across runs is:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ is the overall mean accuracy across all runs.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify agreement across runs, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979) using a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ is the variance between items and σ_{within}^2 is the residual variance across runs. ICC values above 0.75 indicate good reliability, while values above 0.9 imply excellent consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

We also assess dataset stability by testing whether item difficulty remains consistent across runs.

K.2.1 RESAMPLING STUDY

To further assess **robustness**, we perform a resampling analysis. From the total set of $C = 30$ subcategories, we randomly sample subsets of size $S \in 20, 25$ for each subcategory from each of the $R = 3$ runs, across both models and evaluation types (MCQ and open-ended).

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

The mean accuracy across runs for each subset size is:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

and the corresponding standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary nonparametric check, the Wilcoxon signed-rank test measures median differences between paired samples. Let $d_r = A_{r,20} - A_{r,25}$ be the difference in accuracies for run r , excluding ties ($d_r \neq 0$). The test statistic is computed as:

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of signed ranks of positive and negative differences, respectively. The standardized z -score is given by:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

If the resulting p -value satisfies $p > 0.05$, we fail to reject H_0 , indicating that the accuracies from 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , its difficulty can be defined as $d_i = 1 - \bar{a} * i$. If item difficulty is consistent, the correlation between runs should be high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) indicates stable item difficulty and reliable dataset behavior.

K.2.3 CRONBACH'S ALPHA

To measure overall internal consistency, we compute Cronbach's alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 is the variance of run r and s_T^2 is the variance of the total scores aggregated across runs. Values of $\alpha \geq 0.9$ indicate excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We perform the analysis on two models: one large proprietary model, **Gemini-2.5-Flash**, and one small open-source model, **Qwen-2.5-vl-7b-Instruct**, thereby covering both large- and small-scale models as well as proprietary and open-source paradigms. We also conduct all tests in both multiple-choice (MCQ) and open-ended evaluation formats (treated as binary: correct or incorrect). For statistical analysis, we used the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for wilcoxon test; `pingouin`³ (Vallat, 2018) for intraclass correlation and `cronbach_alpha`; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We analyzed the stability of model accuracy across $R = 3$ independent runs. The standard deviation of accuracy (σ_A) across runs is negligible ($< 0.6\%$), indicating that the reported scores are precise estimates unaffected by random seeding noise.

Table 26: Model accuracy and variance across three independent runs.

Model	Format	Mean Accuracy (A)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The consistently low standard deviations across all models and evaluation formats indicate that performance estimates are highly stable and reproducible, with minimal sensitivity to stochastic run-to-run variation. This stability suggests that observed accuracy differences meaningfully reflect underlying model capability rather than noise introduced by random initialization or decoding. Notably, although open-ended evaluation exhibits slightly higher variance than MCQ, the absolute magnitude remains negligible, confirming that free-form generation is also robustly assessed under the current protocol. These findings imply that the dataset size and scoring methodology are sufficient to yield convergent accuracy estimates without requiring extensive repeated runs. Consequently, single-run evaluations are likely to be reliable for large-scale comparisons, while multi-run analyses primarily serve to validate robustness. Overall, this strengthens the empirical credibility of the reported results and supports confident cross-model and cross-format performance interpretation.

Intra-Class Correlation (ICC)

We computed ICC(3,k) to quantify the reliability of the scoring mechanism.

Table 27: Intra-Class Correlation (ICC) reliability scores across repeated runs.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The consistently high ICC(3,k) scores demonstrate that the evaluation exhibits near-perfect reliability across independent runs, regardless of model size or response format. This indicates that item-level difficulty ordering is preserved, and that performance fluctuations do not alter relative rankings among samples. Such strong agreement suggests that the scoring pipeline is internally consistent and not sensitive to stochastic variation in model outputs. Importantly, this rules out random guessing as a dominant factor, even in open-ended generation, where variability is typically higher. From

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

an evaluation standpoint, these results imply that the benchmark provides a dependable signal for assessing model capability. Consequently, longitudinal comparisons and fine-grained ablations can be conducted with high confidence in measurement stability.

Overall, the combined evidence from multiple-run variance analysis and ICC reliability demonstrates that the evaluation is highly robust to stochastic effects. Both absolute performance (low standard deviation) and relative performance (near-perfect ICC) remain stable across independent runs, confirming that results are not driven by random seeding or decoding noise. This robustness ensures that observed performance gaps across models and formats reflect genuine capability differences. As a result, the evaluation framework provides a reliable and reproducible basis for comparative analysis and benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To ensure statistical reliability, this resampling procedure is repeated over **1,000 independent trials**, each time drawing new random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the distribution of accuracy differences and p -values, ensuring that conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials therefore provides strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling study over 1,000 trials using the Wilcoxon signed-rank test ($S = 20$ vs. $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-v1-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-v1-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results indicate that accuracy estimates have converged even at relatively small subcategory sizes. The consistently high mean p -values and the absence of statistically significant differences across all 1,000 trials show that performance metrics are insensitive to moderate sample-size reductions. This stability holds across models and evaluation formats, suggesting that neither model scale nor output format introduces additional sampling uncertainty. The results imply that accuracy variance is dominated by model behavior rather than subcategory-level data sparsity. Consequently, the current subcategory size (≈ 50) provides a safety margin beyond the minimum needed for reliable estimation, supporting the benchmark design and indicating that future extensions could reallocate samples without compromising statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We measure the mean pairwise Pearson correlation ($\bar{\rho}$) between runs and Cronbach’s Alpha to assess internal consistency.

Table 29: Item-level consistency metrics across repeated runs.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-v1-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-v1-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The high mean pairwise correlations (> 0.94) demonstrate strong item-level stability, indicating that the relative difficulty of individual questions is preserved across independent runs. This consistency shows that model responses are systematically structured rather than dominated by random variation, even under stochastic decoding. The high Cronbach’s α values (> 0.9) further indicate excellent internal coherence, with items contributing reliably to the aggregate accuracy measure. The slight reductions observed for open-ended formats are expected because the response space is more flexible, but the values remain within the excellent-reliability range. Together, these results indicate that the benchmark operates as a unified and internally consistent measurement instrument, so performance differences can be attributed to capability variation rather than item-level instability.

K.3.3 OVERALL DISCUSSION

The expanded statistical tests show that the evaluation protocol and dataset are robust. **ICC and Cronbach’s α scores consistently exceed 0.98**, indicating near-perfect reliability and internal consistency. The **standard deviation of accuracy across runs remains negligible** ($< 0.6\%$), confirming that reported model performance is stable with respect to random seeding and decoding variability. The **resampling study** further shows that smaller subsets ($N = 20$ and $N = 25$) produce statistically indistinguishable accuracy estimates relative to the full set, establishing that the chosen subcategory size (~ 25) is sufficient for convergence. Collectively, absolute scores and relative rankings are stable across runs and sampling regimes. Thus, the fine-grained comparisons across models, formats, and subcategories reflect genuine capability differences rather than sampling variance or evaluation noise.